MAHIDOL UNIVERSITY
Faculty of Information and Communication Technology

Lecture 7
Transmission Media (Part 1)

September 2026

มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

Unless allowed by Mahidol University, the User shall not copy, modify, or exploit in part or in whole of the copyrighted materials on the Platform, other than for your own individual study. The copy, modify or communication to public of the copyrighted materials shall be inflicted with a fine from 20,000 Baht up to 200,000 Baht. If the offence is committed with the commercial purpose, the offender shall be inflicted with imprisonment for a term from 6 months up to 10 years or a fine from 100,000 Baht up to 800,000 Baht or both imprisonment and fine.

Quotes of the Day

Mahidol University

What you think,
you become.

What you feel,
you attract.

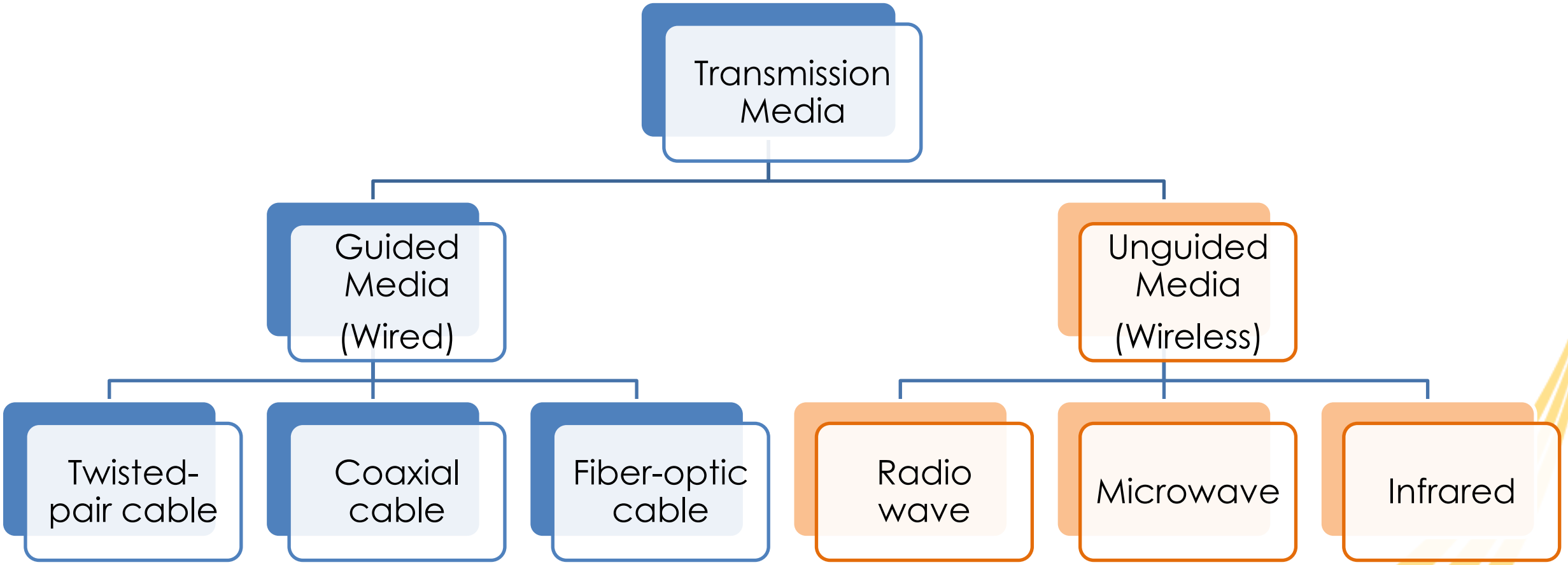
What you imagine,
you create.

-BUDDHA

"All your dreams can come true if we have the courage to pursue them."
— Walt Disney



Transmission Media



Guided Media (Wired)

- Guided media is usually media that guide signals to travel along the path of the medium.
- A signal traveling along any of these media is directed and contained by the **physical limits of the medium**



Twisted-pair cable



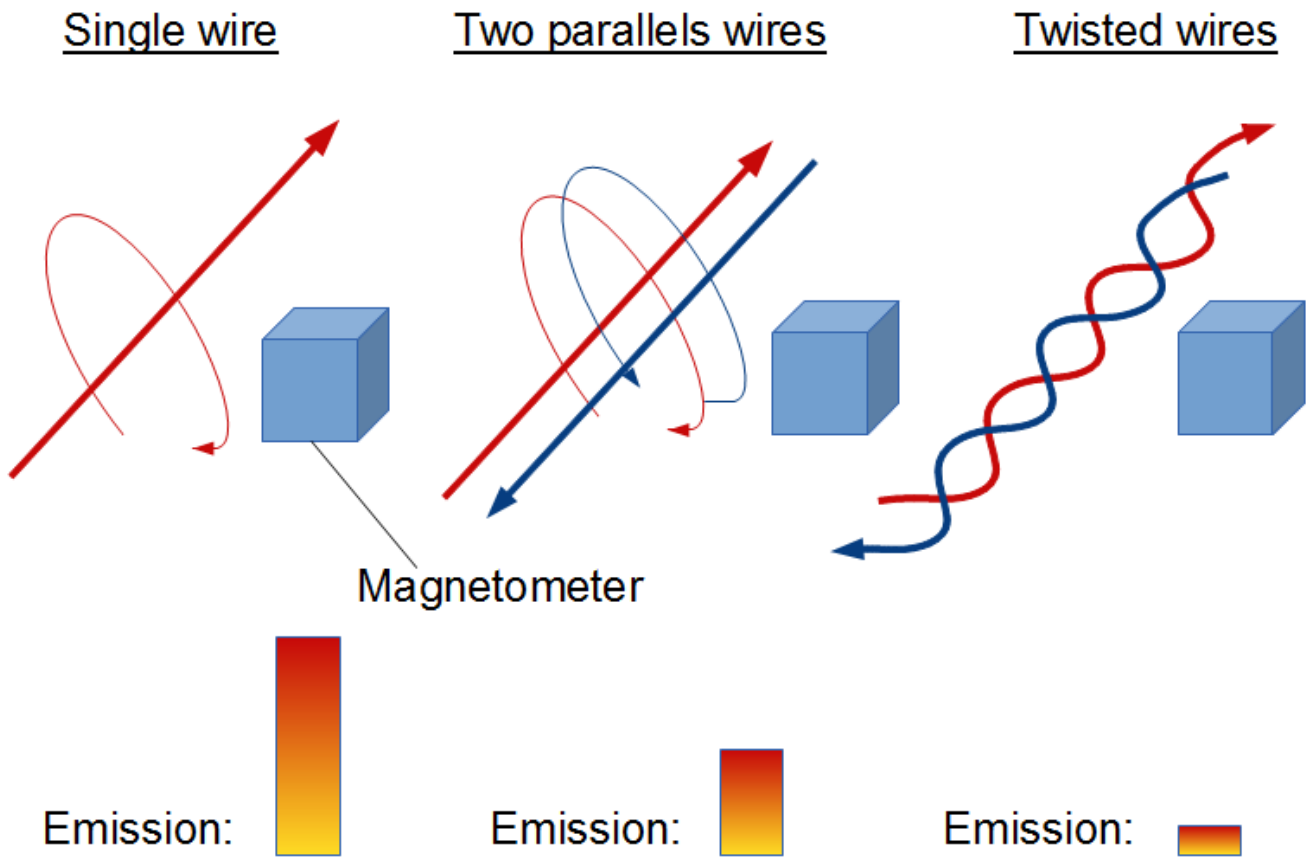
Coaxial cable



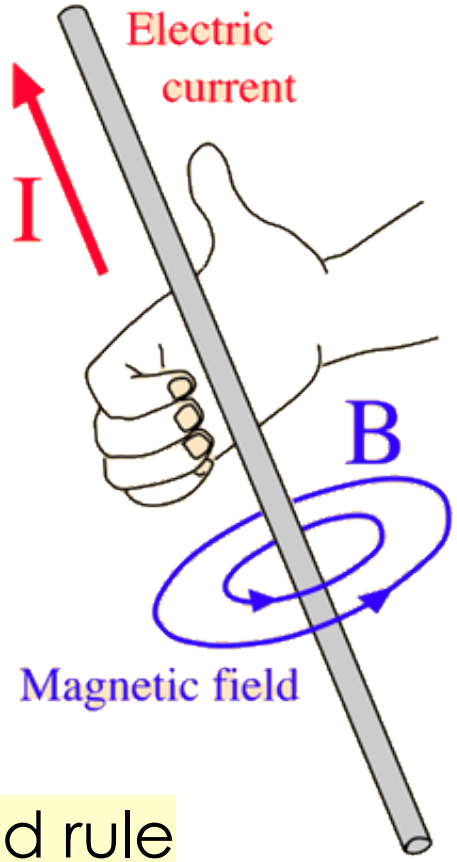
Fiber-optic cable

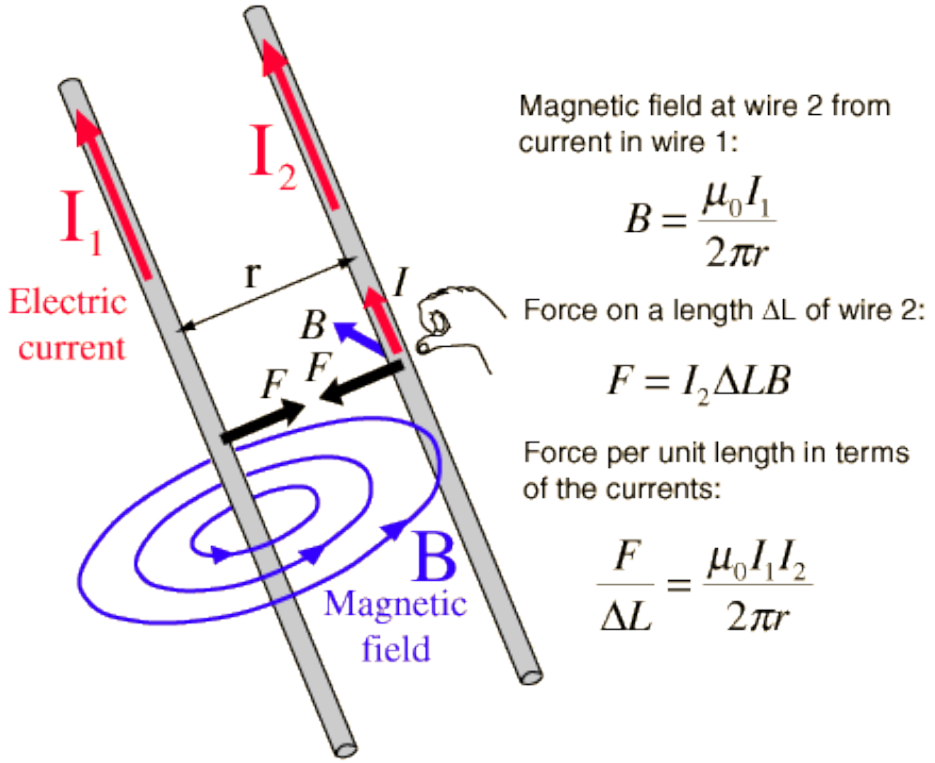
Twisted-pair Cable

- Two conductors (normally **copper**) are twisted together in order to reduce noise or crosstalk.



Crosstalk





Magnetic field at wire 2 from current in wire 1:

$$B = \frac{\mu_0 I_1}{2\pi r}$$

Force on a length ΔL of wire 2:

$$F = I_2 \Delta L B$$

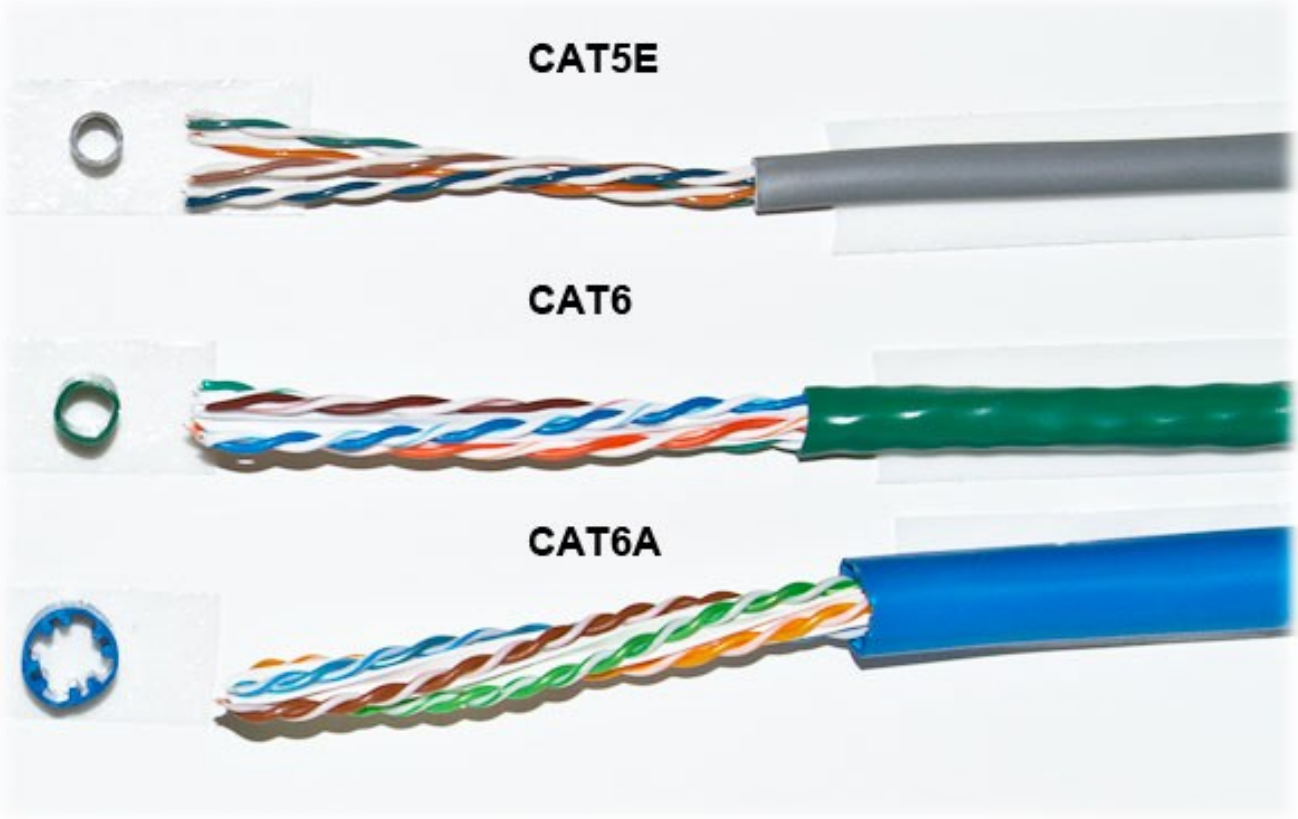
Force per unit length in terms of the currents:

$$\frac{F}{\Delta L} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

the right-hand rule

Twisting the Cable

- The number of twists per unit length determines the quality of the cable.
- The more twists, the better quality.



What are the differences between CAT6, CAT6e, and CAT6A?





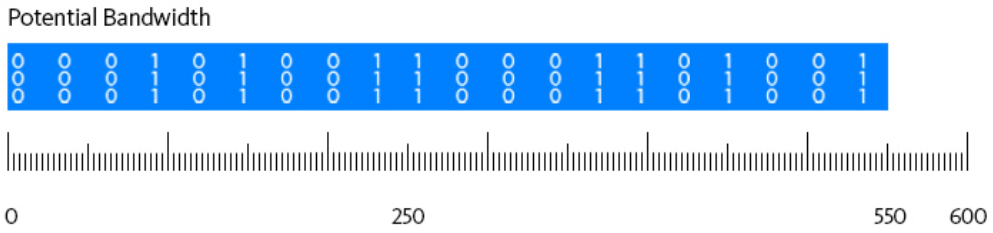


https://www.cablesys.com/updates/cat6-cat6e-cat6a-differences/

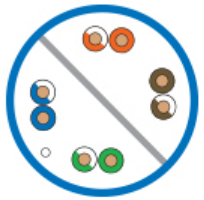
CAT6, CAT6A, CAT6e



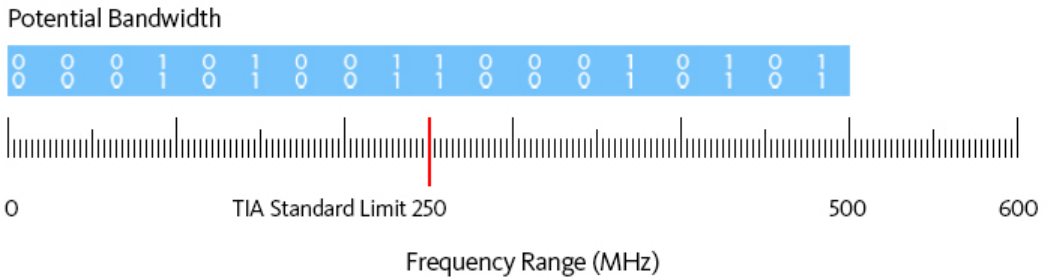
CAT6e



there is no CAT6e standard, but proposed by vendors.

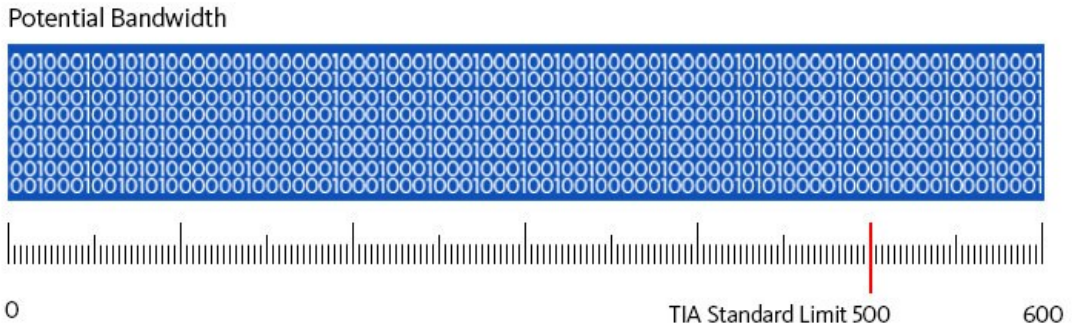


CAT6





CAT6A



https://cablesys.com/updates/cat6-cat6e-cat6a-differences/

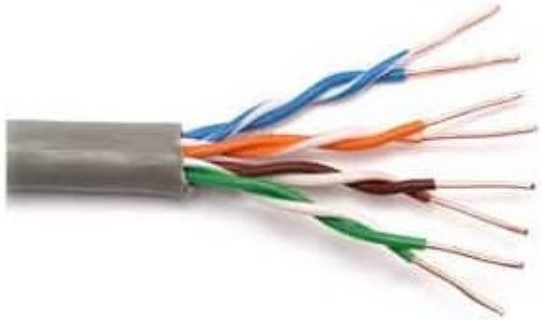
CAT6, CAT6A, CAT6e

Common Usage	CAT6	CAT6e	CAT6A
Phone Lines	✓	✓	X
Home Network	✓	✓	X
Office Network	✓	✓	✓
Data Center	X	✓	✓
Features/Specs			
Potential Bandwidth (per sec)	1Gb	1Gb	10Gb
Time to transfer 1 Terabyte	3 hours	3 hours	20 minutes
Data Transmission	1000 BASE-TX	Exceeds 1000 BASE-TX	10GBASE-T
Connector Type	RJ45 8P8C (for CAT6)	RJ45 (for CAT6)	RJ45 (for CAT6A)
Frequency Range Minimum	0 - 250 MHz	0 - 250 MHz	0 - 500 MHz
Frequency Maximum	500 MHz	550 MHz	600 MHz
Performance Distance	328 Feet	328 Feet	328 Feet
Alt. Distance	10Gb 180ft	10Gb 180ft	-

https://cablesys.com/updates/cat6-cat6e-cat6a-differences/

Unshielded vs Shielded Twisted-pair Cable

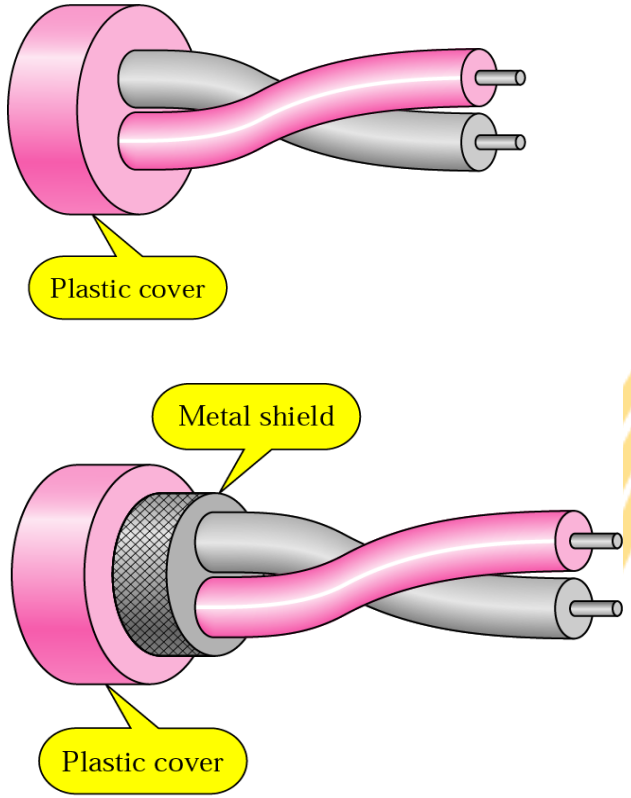
- Shielded twisted-pair (STP) cable has an additional magnetic field protection layer over unshielded twisted-pair (UTP) cable in noisy environment such as heavy industry factory or hospital.
- **STP**
  - Produced by **IBM**
  - Prevent the penetration of noise and crosstalk
  - Bulkier and more expe
  - Hard to bend



UTP Cable



STP Cable



Categories of Twisted-pair Cables

- **Electronic Industries Association (EIA)** has defined standards to classify twisted-pair cable into categories.

UTP Category	Data Rate	Max. Length	Cable Type	Application
CAT1	Up to 1Mbps	-	Twisted Pair	Old Telephone Cable
CAT2	Up to 4Mbps	-	Twisted Pair	Token Ring Networks
CAT3	Up to 10Mbps	100m	Twisted Pair	Token Rink & 10BASE-T Ethernet
CAT4	Up to 16Mbps	100m	Twisted Pair	Token Ring Networks
CAT5	Up to 100Mbps	100m	Twisted Pair	Ethernet, FastEthernet, Token Ring
CAT5e	Up to 1 Gbps	100m	Twisted Pair	Ethernet, FastEthernet, Gigabit Ethernet
CAT6	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT6a	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT7	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (100 meters)

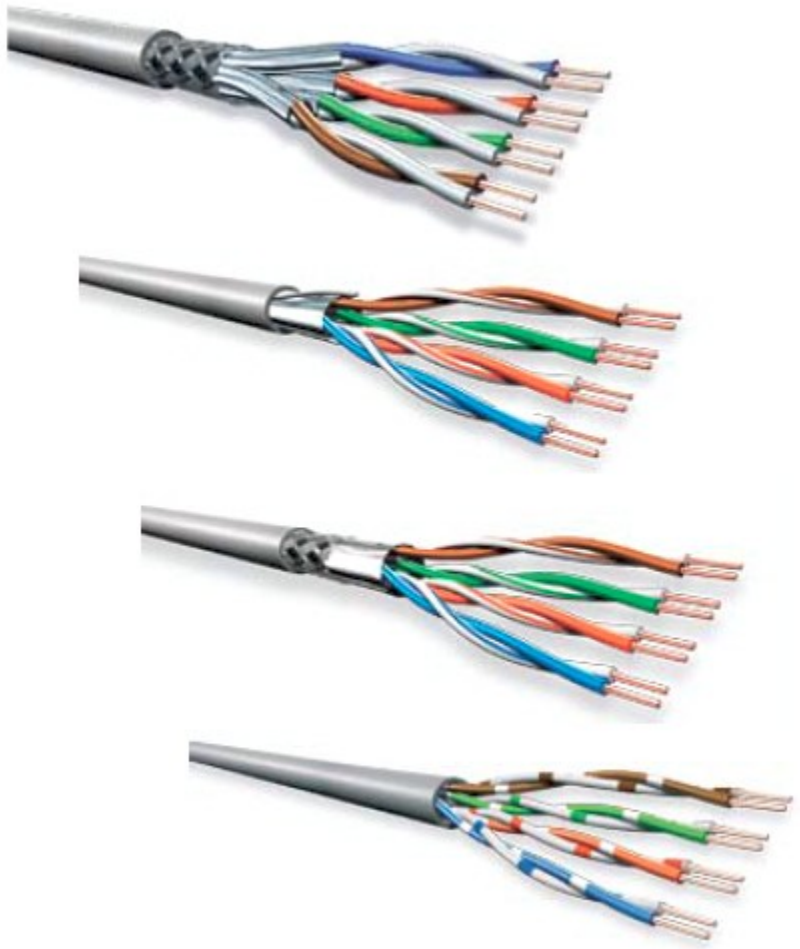
CAT



Categories of Twisted-pair Cables

Name	Typical construction	Bandwidth	Applications	Notes
Level 1		0.4 MHz	Telephone and modem lines	Not described in EIA/TIA recommendations. Unsuitable for modem systems.
Level 2		4 MHz	Older terminal systems, e.g. IBM 3270	Not described in EIA/TIA recommendations. Unsuitable for modem systems.
Cat.3	UTP	16 MHz	10BASE-T and 100BASE-T4 Ethernet	Described in EIA/TIA-568. Unsuitable for speeds above 16 Mbit/s. Now mainly for telephone cables
Cat.4	UTP	20 MHz	16 Mbit/sToken Ring	Not commonly used
Cat.5	UTP	100 MHz	100BASE-TX & 1000BASE-T Ethernet	Common in most current LANs
Cat.5e	UTP	100 MHz	100BASE-TX & 1000BASE-T Ethernet	Enhanced Cat5. Same construction as Cat5, but with better testing standards.
Cat.6	UTP	250 MHz	10GBASE-T Ethernet	Most commonly installed cable in Finland according to the 2002 standard. SFS-EN 50173-1
Cat.6a	U/FTP, F/UTP	500 MHz	10GBASE-T Ethernet	Adds cable shielding. ISO/IEC 11801:2002 Amendment 2.
Cat.7	F/FTP, S/FTP	600 MHz	Telephone, CCTV, 1000BASE-TX in the same cable. 10GBASE-T Ethernet.	Fully shielded cable. ISO/IEC 11801 2nd Ed.
Cat.7a	F/FTP, S/FTP	1000 MHz	Telephone, CATV, 1000BASE-TX in the same cable. 10GBASE-T Ethernet.	Uses all four pairs. ISO/IEC 11801 2nd Ed. Am. 2.
Cat.8.1	U/FTP, F/UTP	1600-2000 MHz	Telephone, CATV, 1000BASE-TX in the same cable. 40GBASE-T Ethernet.	In development.
Cat.8.2	F/FTP, S/FTP	1600-2000 MHz	Telephone, CATV, 1000BASE-TX in the same cable. 40GBASE-T Ethernet.	In development.

Extra Protection



Different types of twisted pair cables

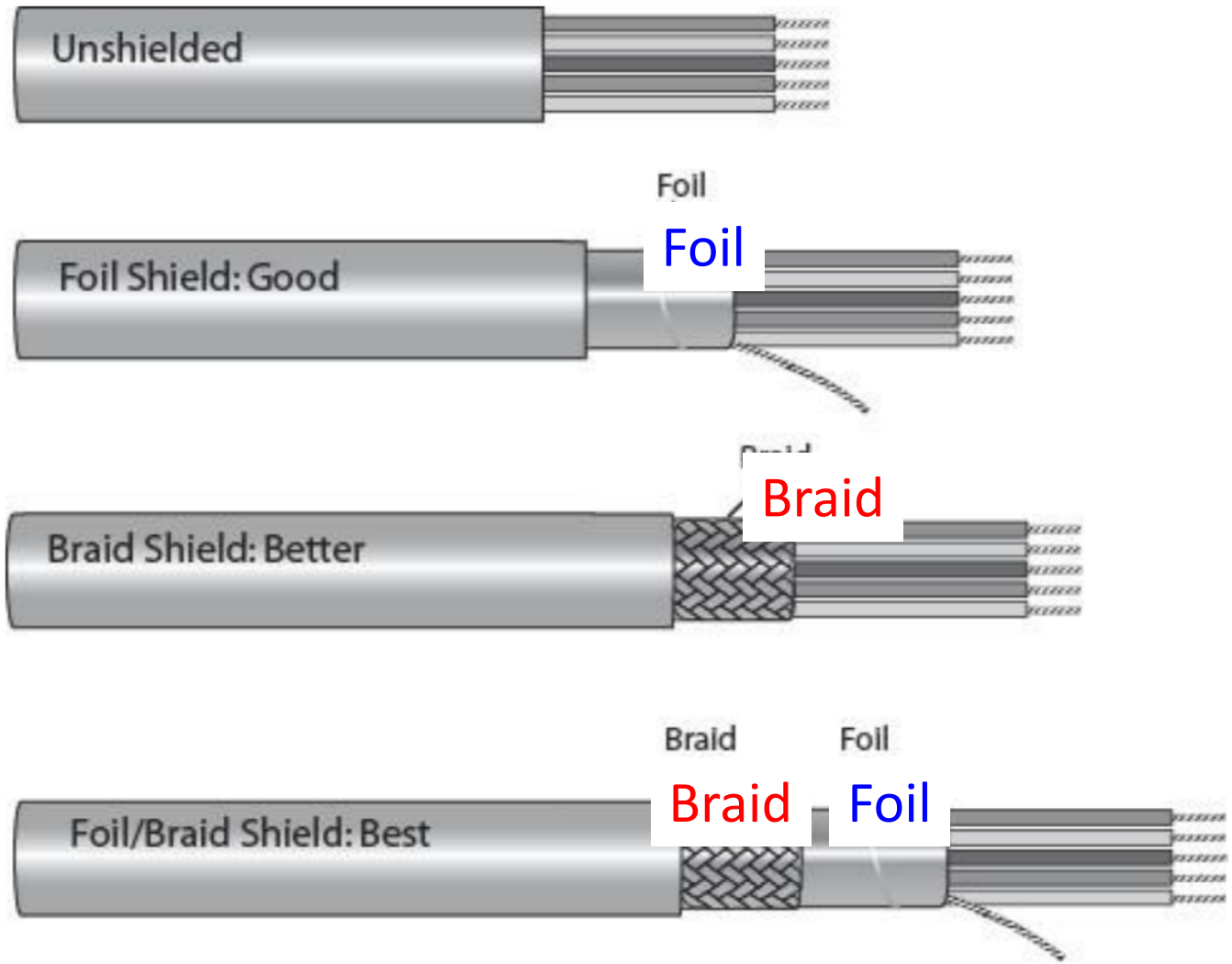
S/FTP:
overall braid screen (S),
elements foil screened (FTP)

F/UTP:
overall foil screen (F),
elements unscreened (UTP)

SF/UTP:
overall braid and foil screen (SF),
elements unscreened (UTP)

U/UTP:
no overall screen (U),
elements unscreened (UTP)

Extra Protection



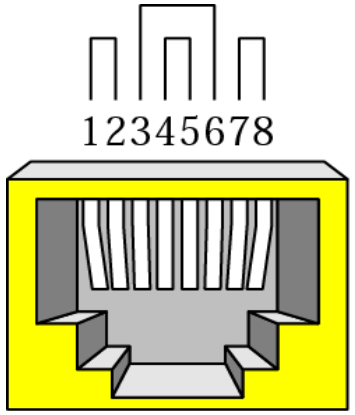
Weaving metal

Note

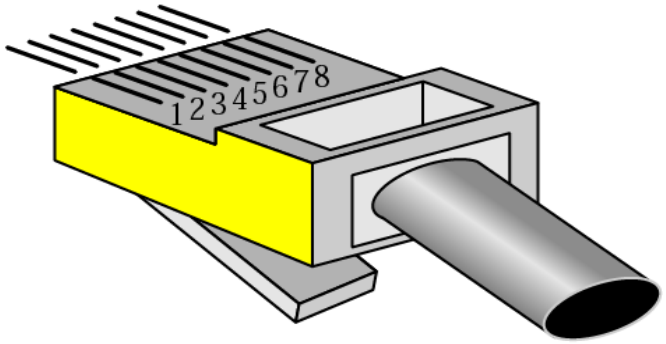
- Make sure you have a cable with sufficient shielding for the application's needs. In moderately noisy environments, a foil alone may provide adequate protection. In noisier environments, consider braids or foil-braid combinations.
- Use a cable suited to the application. Cables that experience repeated flexing usually use a spirally wrapped shield rather than a braid. Avoid foil-only shielding on flex cables since continuous flexing can tear the foil.
- Make sure the equipment that the cable is connected is properly grounded. Use an earth ground wherever possible and check the connection between the ground point and the equipment. Eliminating noise depends on a low resistance path to ground.

Twisted-pair Cable Connectors

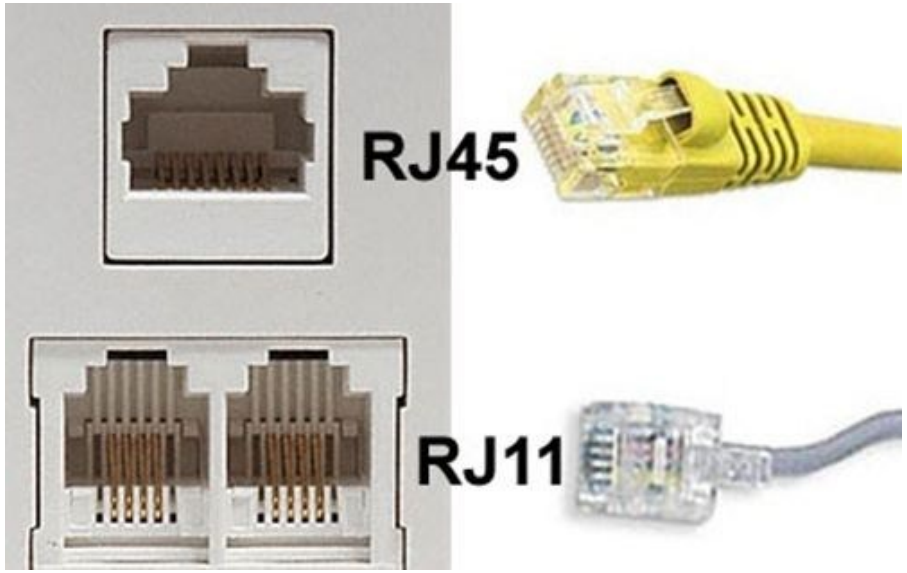
- Registered Jack (RJ)



RJ-45 Female



RJ-45 Male



LAN cable

Telephone line

Connecting RJ-45 Connector to Cable

U.S. Government regulations require the use of the preferred TIA/EIA 568A in federal contracts.

TIA/EIA 568B the most widely used

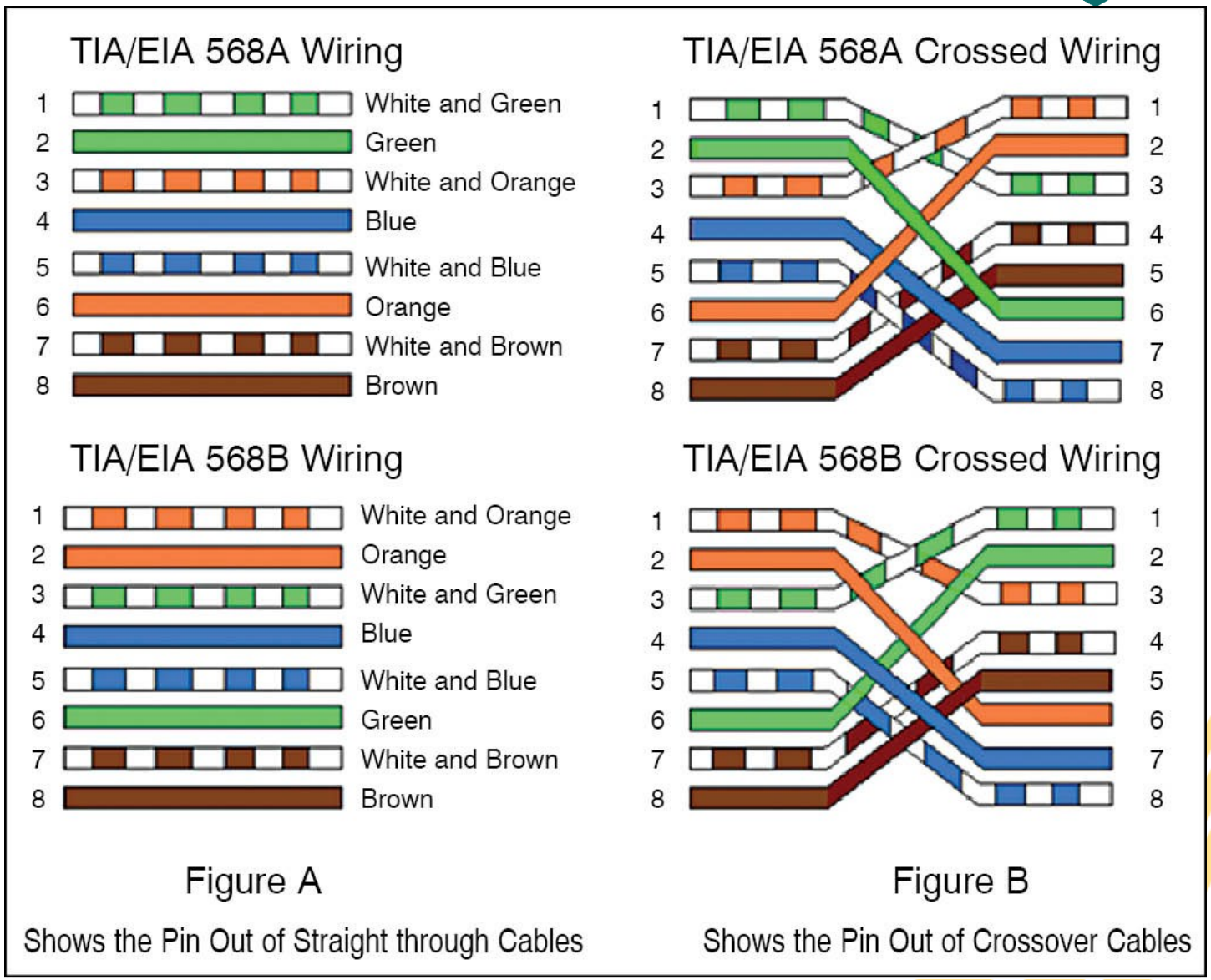


Figure A

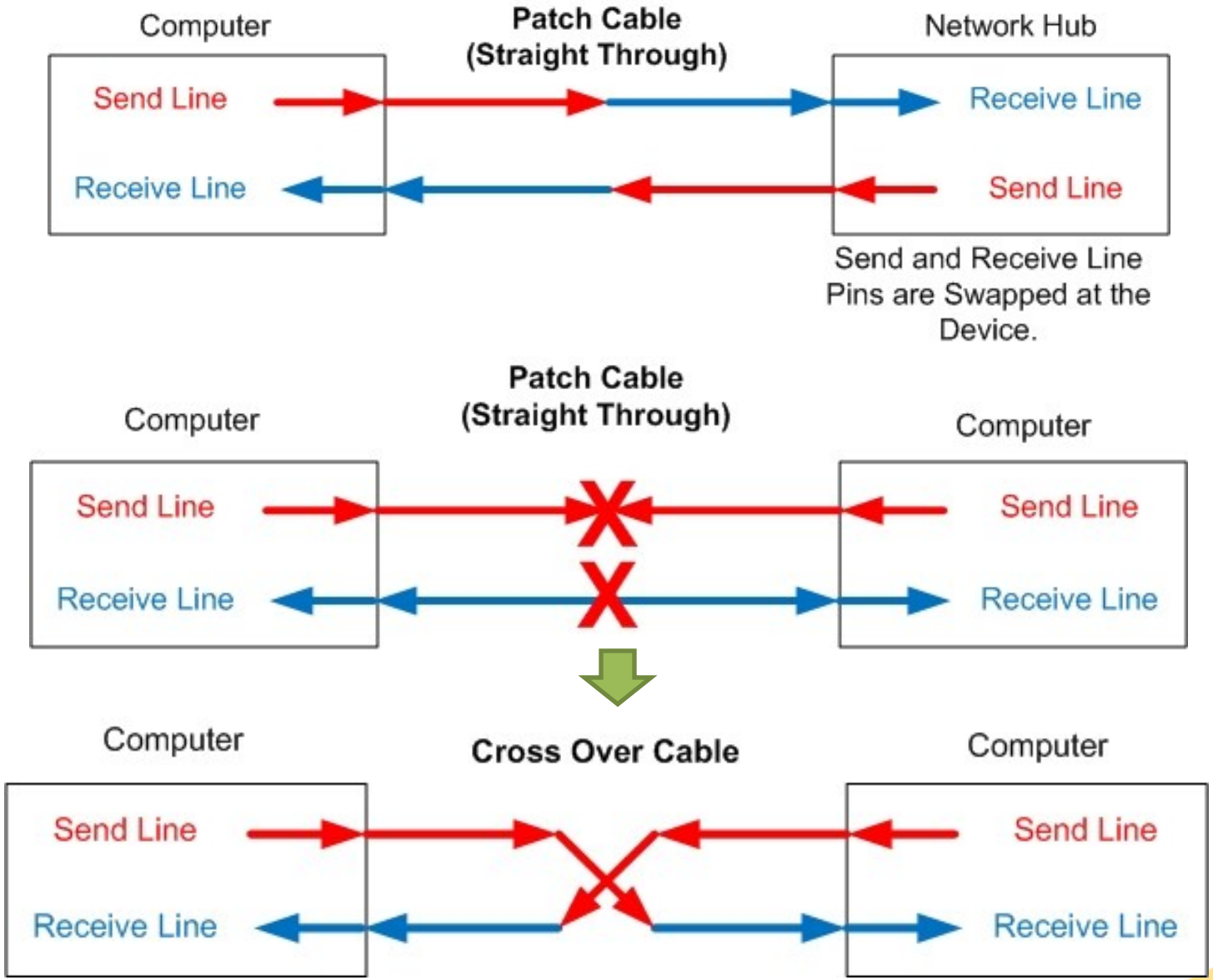
Shows the Pin Out of Straight through Cables

Figure B

Shows the Pin Out of Crossover Cables

https://www.networktechinc.com/standard.html

Patch vs Crossover Cables



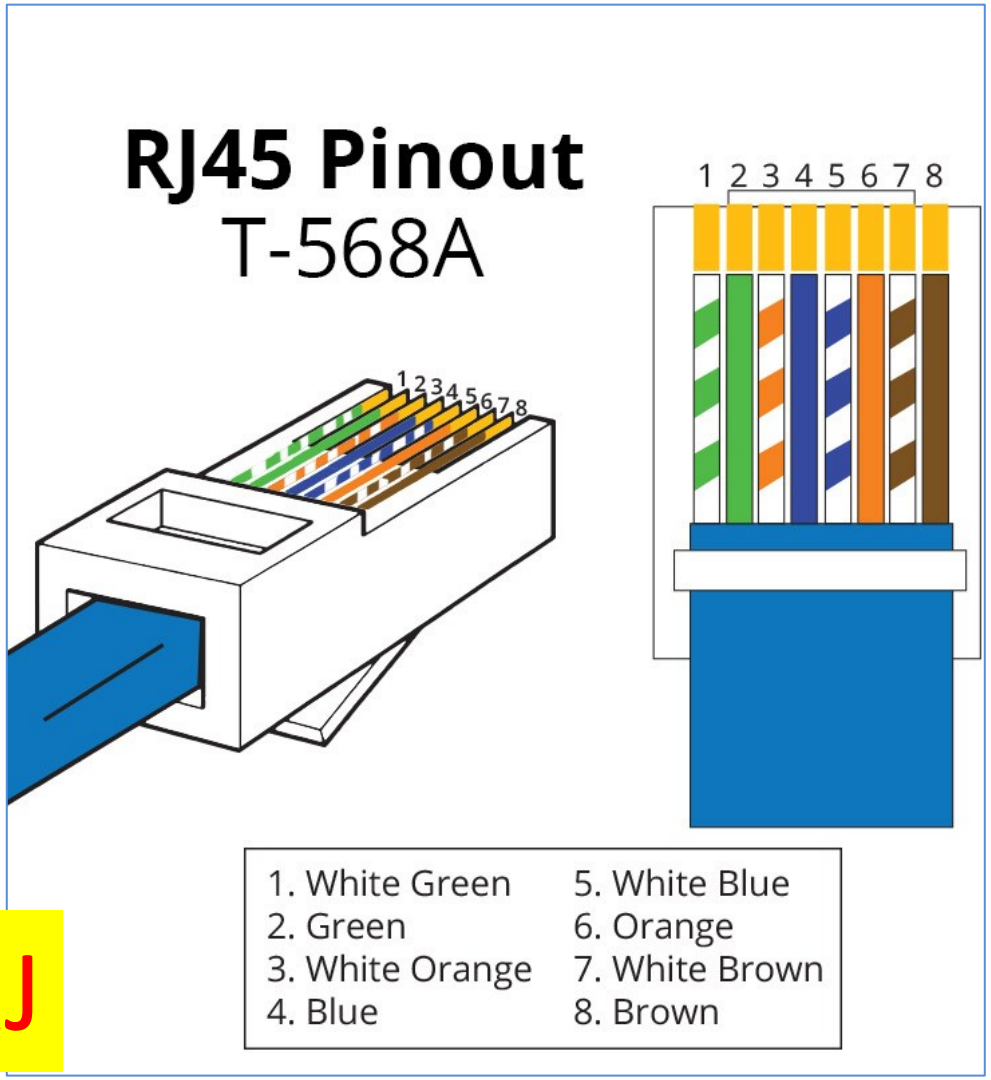
Patch Cable vs Crossover Cable

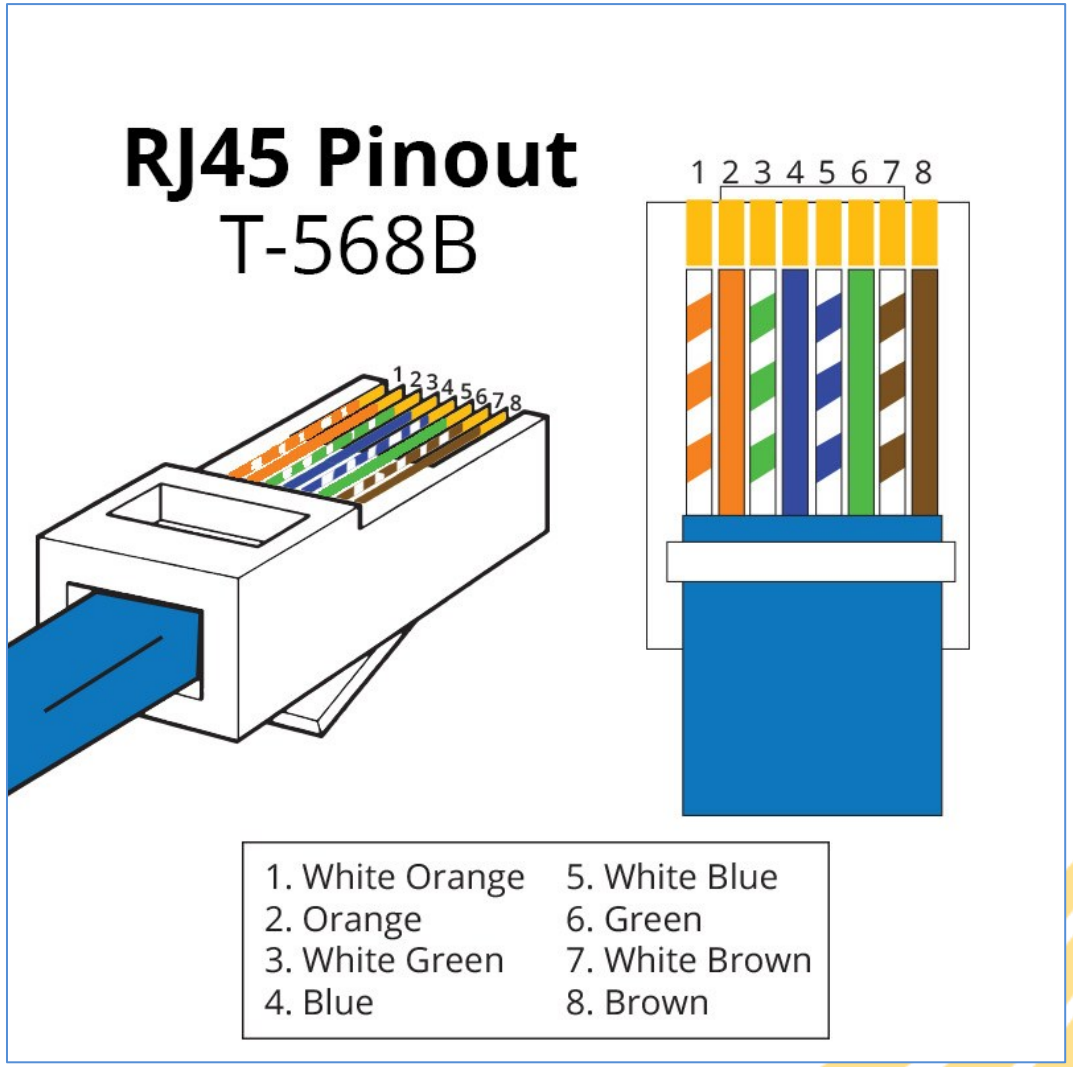
- **A patch cable is straight through and is what is used to connect between different types of devices** (computer to network switch).

- A **crossover cable** connects the transmit pair to the receive pair at the other end (and vice versa). This allows two devices (computers) to talk to each other directly. If you were to plug two computers together with a straight through (patch) cable, they'll both be talking and listening on the same wire and nothing will work. A crossover cable has it's input and output lines crossed over so that you can connect 2 computers directly to one another without a router or hub or switch between them. → **connects same types of devices**
  - Odds are you will never need a crossover cable unless you are working with commercial grade networking equipment that sometimes use them for terminal connections.

- **A switch/hub/modem** generally knows to send/receive on the other pairs, and so transmits to two devices connected to it correctly. Other uses for crossover cables is to connect two switches (though many **auto-sense** that situation these days or have an uplink port that is already crossed over).

Connecting RJ-45 Connector to Cable

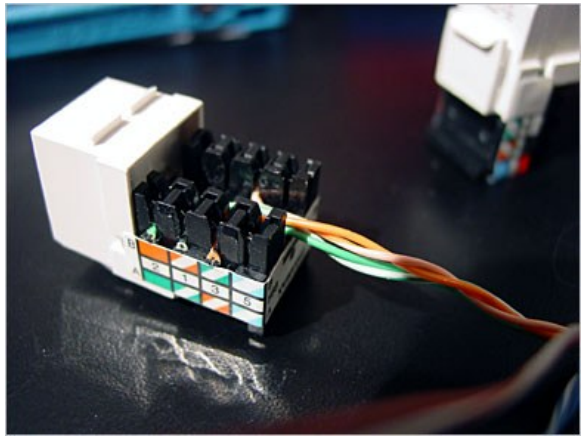


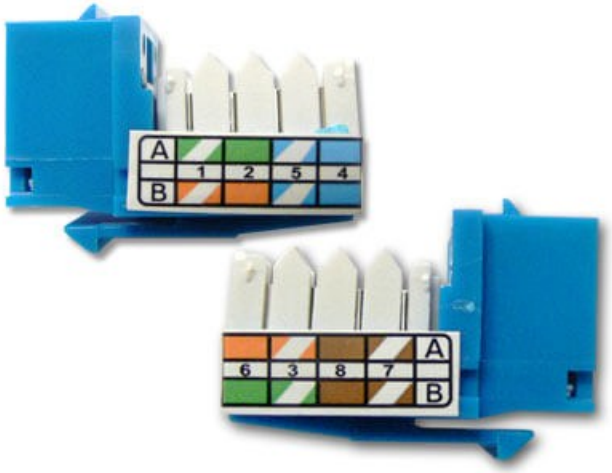


RJ

Connecting RJ-45 Connector to Cable

- Female jack usually have labels (T-568A and T-568B)





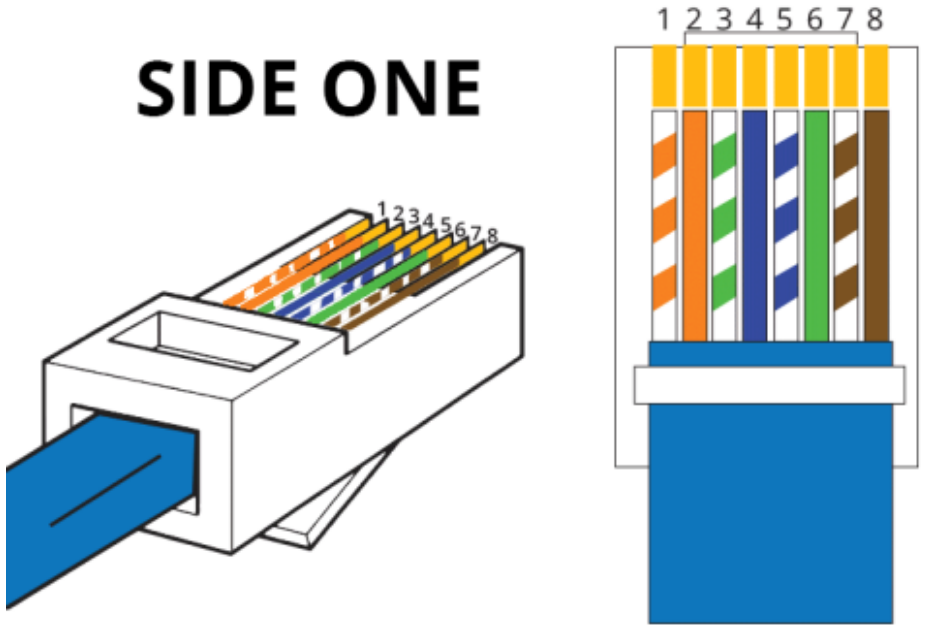




Connecting RJ-45 Connector to Cable

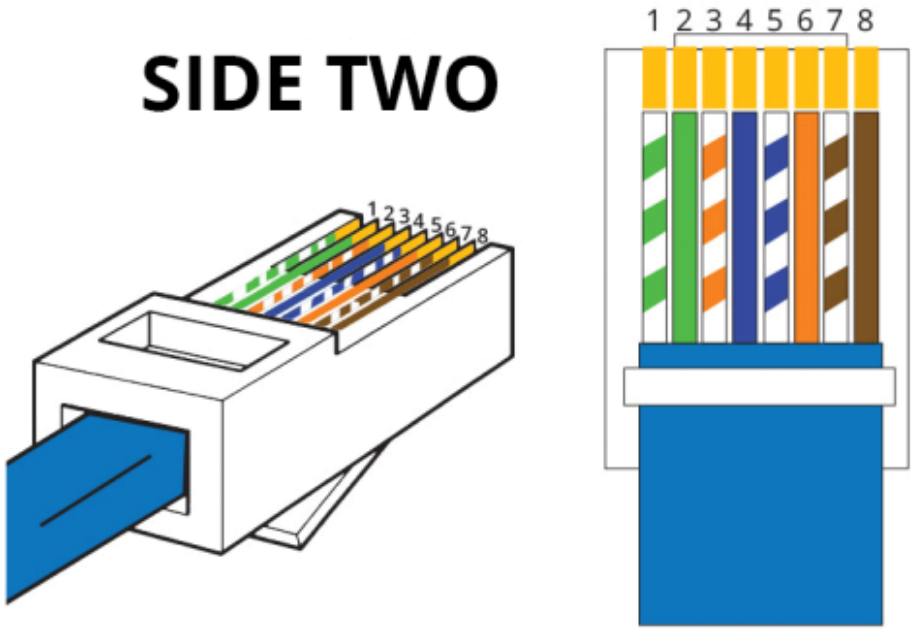
CROSSOVER PINOUT

SIDE ONE



1. White Orange	5. White Blue
2. Orange	6. Green
3. White Green	7. White Brown
4. Blue	8. Brown

SIDE TWO



1. White Green	5. White Blue
2. Green	6. Orange
3. White Orange	7. White Brown
4. Blue	8. Brown

Cabling 101

1. Read cable specification and document to find out whether you use the cable with sufficient quality that fits your need (speed and reliability performance) within your budget.
2. Cables may be taken out and added all the time. You may want to use the cables that have easy-to-use connectors and flexible easy-to-access wiring path (conduits)
3. Always know where to wire cables and have cabling plan. In some situation, cables have to be bended. You have to pick flexible cables which may reduce the quality of the cable.
4. Do not put data cables near power cables and heavy machinery.
5. Do not wire cable in hazardous environment (wet, animals, people walking pass)
6. Use different colors and labels of cables (priority, importance, paths)
7. Size and weight of cables also matter when the conduits have limited size. Consider multiplexing.

Cable Tray, Cable Conduit, Cable Pit

Feature	Cable Tray	Cable Conduit	Cable Pit
Visibility	Fully accessible	Fully enclosed	Accessible via the lid
Heat Dissipation	Excellent	Poor	N/A
Typical Environment	Overhead / Indoor	Wall-mount / Outdoor	Underground
Expansion Ease	Very High	Low	Medium



Cable Tray



Cable Conduit



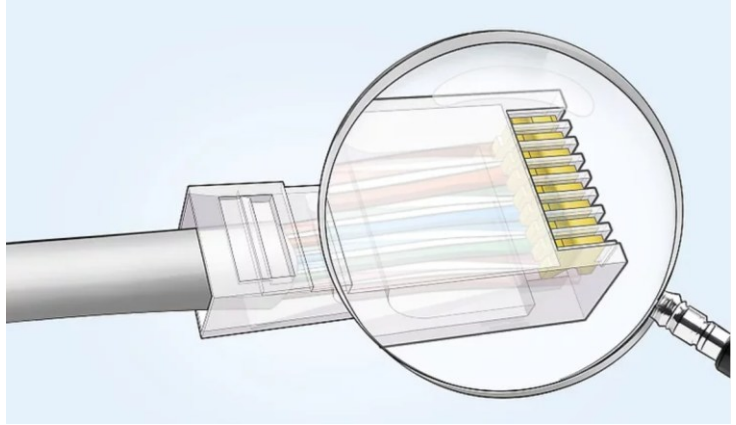
Cable Pit

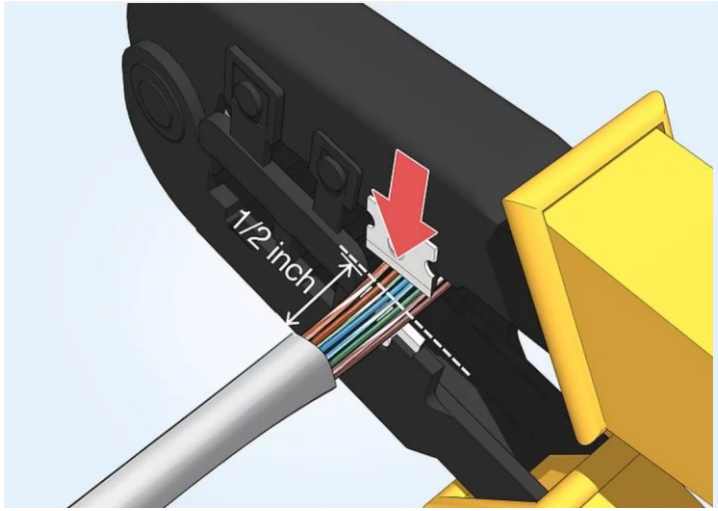
https://www.cabletrayfab.com/cable-pathways-conduits-tray-and-cable-pit/

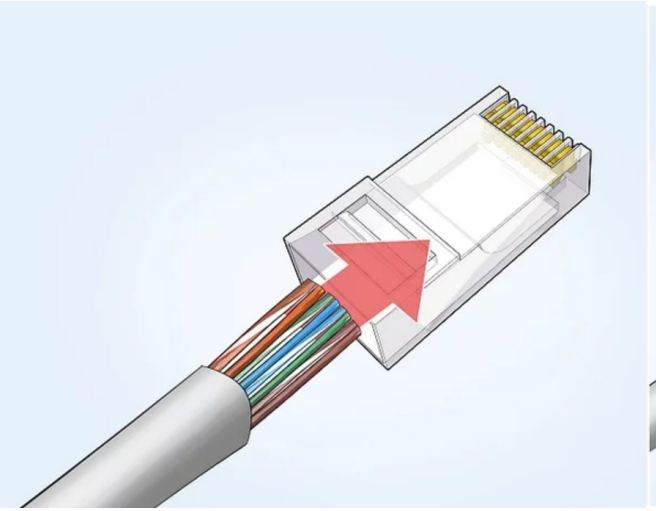
How to Crimp RJ45 to CAT5/CAT6 Cables

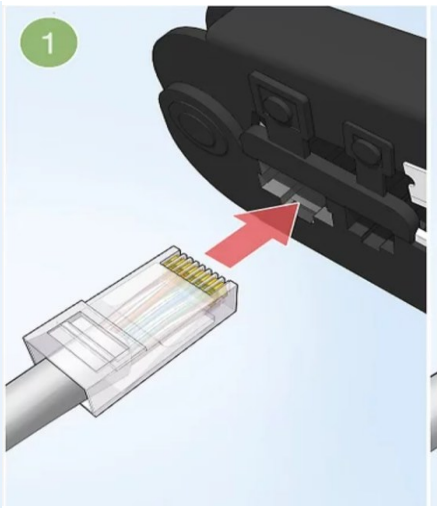
- It is good to be able to crimp standard RJ45 jacks to CAT5/6 cables by yourself.

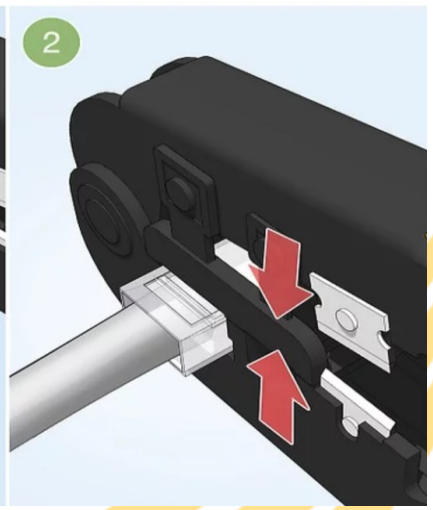
- Reference: https://www.wikihow.com/Crimp-Rj45











RJ-45 Cable Tester

- Test connectivity of each wire inside the CAT 5e or CAT 6 cable

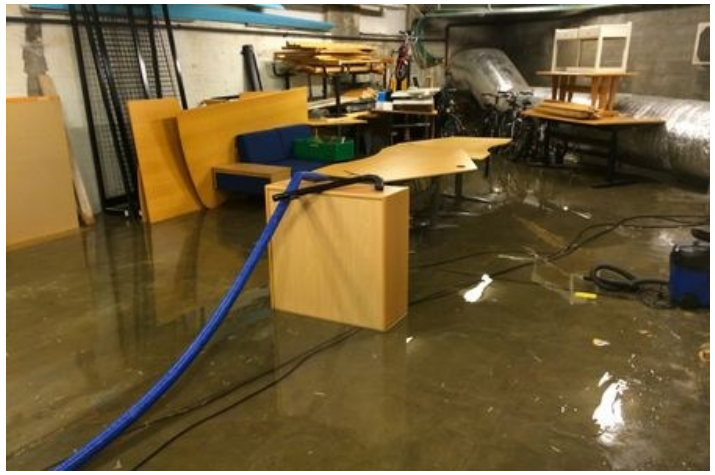


Cable Protection

- Be careful of where you lay your cables!





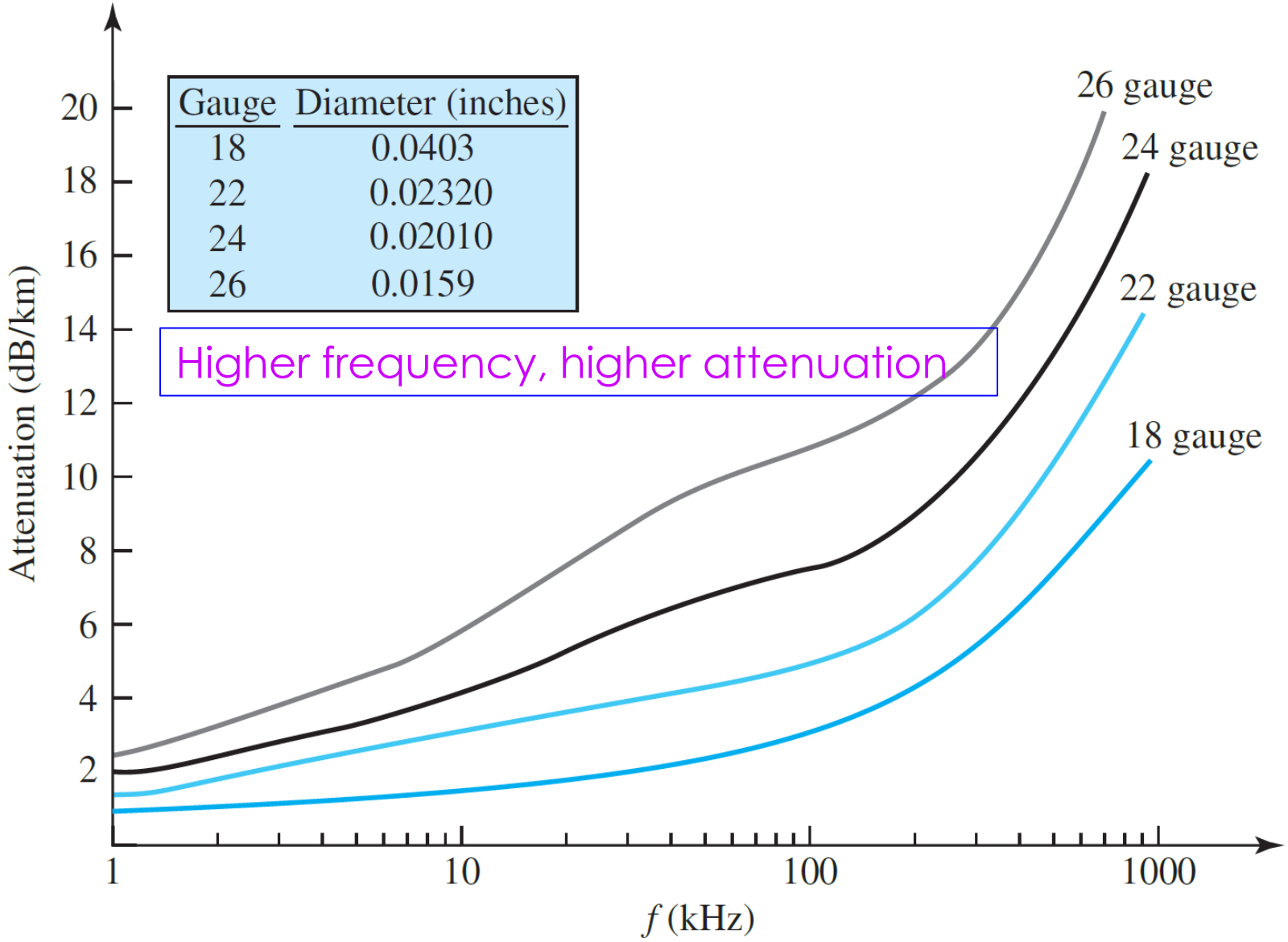




Performance of Twisted-pair Cable

Gauge	Diameter (inches)
18	0.0403
22	0.02320
24	0.02010
26	0.0159

Higher frequency, higher attenuation



Let's do the exercise and have a Break !





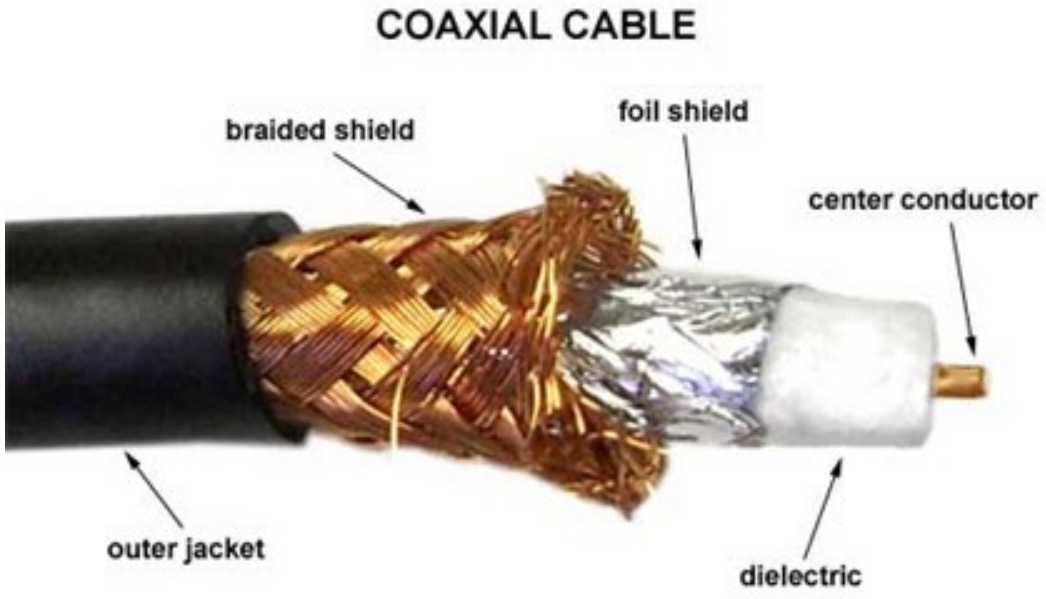


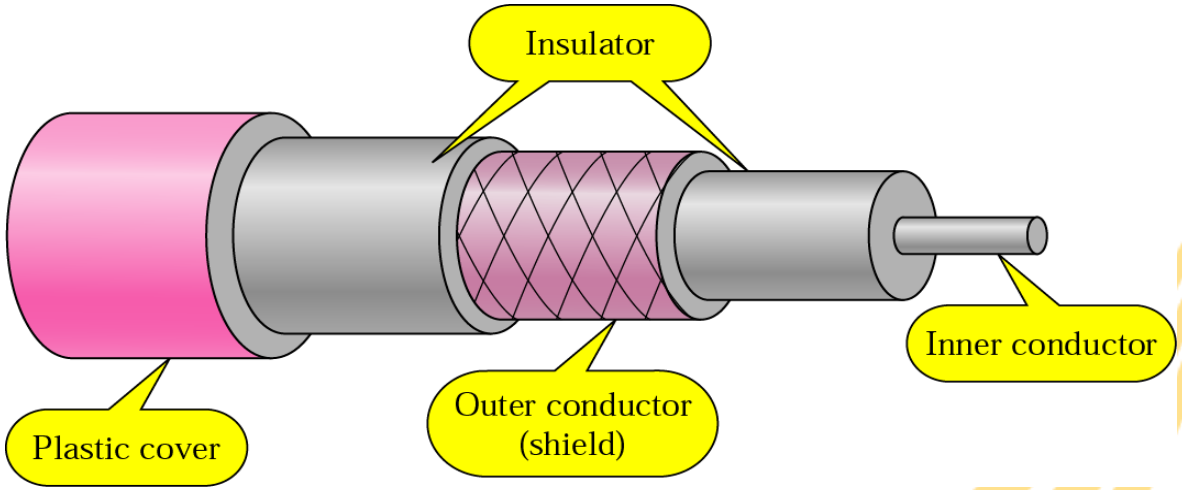




Coaxial Cable

- Carries signals of higher frequency ranges than twisted-pair cable, in part because the two media are constructed quite differently
- The outer metallic wrapping serves both as a shield against noise and as the second conductor.





Types of Coaxial Cable

- There are a number of commonly used coaxial cable in use today.
- Most are "**RG**" (Radio Guide) style cables originally designed for **military applications** but now extensively used in **commercial applications**.

Cable Type	Impedance	Typical Application	Best feature	Trade Off
RG174/U 	50 Ohm	Transmission of data signals in applications such as LAN/WAN or GPS	Small diameter, flexible	Higher signal loss than larger diameter cable such as RG58
RG188A/U 	50 Ohm	Transmission of data signals in applications such as LAN/WAN or GPS in situations where high temperature performance is needed	Small diameter, flexible. High temperature rating of taped TFE outer jacket	Higher signal loss than larger diameter cable such as RG58 and higher cost than standard RG174
RG316/U 	50 Ohm	Transmission of data signals in applications such as LAN/WAN or GPS in situations where high temperature performance is needed	Small diameter, flexible. High temperature rating of extruded FEP outer jacket	Higher signal loss than larger diameter cable such as RG58 and higher cost than standard RG174
RG58C/U 	50 Ohm	Transmission of data signals in applications such as antenna feed cables or Ethernet backbones	Lower signal loss than smaller diameter cable such as RG174	Less flexible than smaller diameter cable such as RG174
RG142B/U 	50 Ohm	Transmission of data signals in applications such as antenna feed cables or Ethernet backbones in situations where high temperature performance is needed	Lower signal loss than smaller diameter cable such as RG174. High temperature rating of extruded FEP jacket	Less flexible than smaller diameter cable such as RG174 and higher cost than RG58C cable.
RG59A/U 	75 Ohm	Transmission of a video or audio signal in applications such as security systems or CATV	Lower signal loss than smaller diameter cable such as RG179. Flexibility of stranded center conductor cable	Higher signal loss than solid center conductor RG59B/U cable

RG

Types of Coaxial Cable

RG59A/U 	75 Ohm	Transmission of a video or audio signal in applications such as security systems or CATV	Lower signal loss than smaller diameter cable such as RG179. Flexibility of stranded center conductor cable	Higher signal loss than solid center conductor RG59B/U cable
RG59B/U 	75 Ohm	Transmission of a video or audio signal in applications such as security systems or CATV	Lower signal loss than smaller diameter cable such as RG179 and RG59A/U stranded center conductor cable	Less flexible than smaller diameter cable such as RG179 or stranded center conductor RG59A/U
RG6/U 	75 Ohm	Transmission of a video or audio signal in applications such as security systems or CATV	Lower signal loss than smaller diameter cable such as RG179 and both RG59A/U or RG59B/U cable	Less flexible than smaller diameter cable such as RG179 and both RG59A/U or RG59B/U
RG223/U 	50 Ohm	Transmission of data signals in applications such as LAN/WAN or GPS in situations where low signal loss and high shielding performance is needed	Lower signal loss and better shielding than smaller diameter cable such as RG174 or RG58C/U cables	Less flexible than smaller diameter cable such as RG174 and higher cost than single shielded RG58C cable
RG213/U 	50 Ohm	Transmission of data signals in applications such as antenna feed cables in situations where low signal loss and high operating voltage performance is needed	Lower signal loss and higher operating voltage than RG58C/U cable	Larger diameter and less flexible than RG58C/U cable
RG179B/U 	75 Ohm	Transmission of a video signal in applications such as security systems where high temperature performance is needed	Small diameter, flexible. High temperature rating of extruded FEP outer jacket	Higher signal loss and cost than larger diameter cable such as RG59
RG187/U 	75 Ohm	Transmission of a video signal in applications such as security systems where high temperature performance is needed	Small diameter, flexible. High temperature rating of TFE taped outer jacket	Higher signal loss and cost than larger diameter cable such as RG59

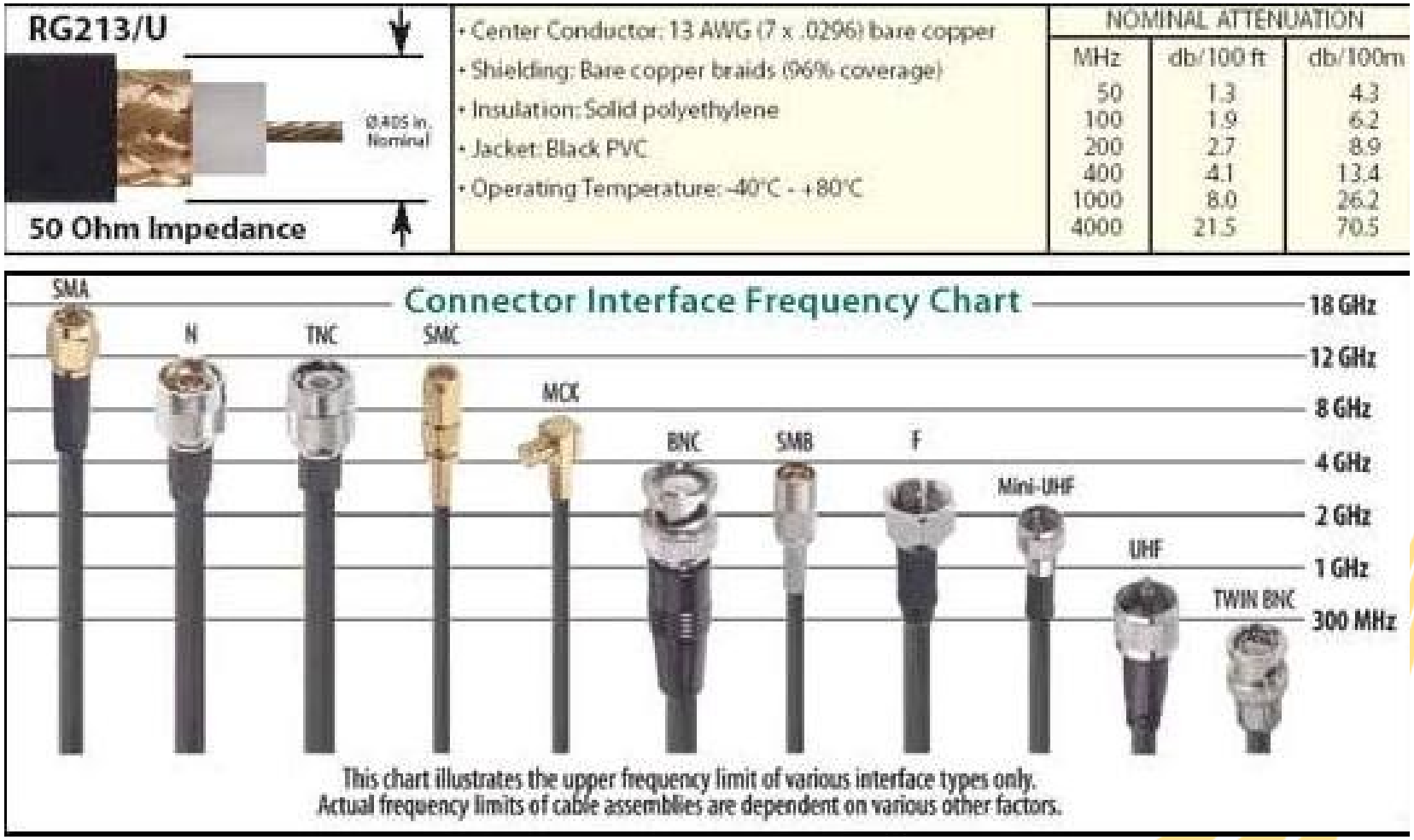
Coaxial Cable Shielding Type

Shielding Type			
Single Braid Shield (95% coverage)	Single Braid Shield (60%) + Foil Wrap (100%)	(2) Braids (60%) + (2) Foil Wraps (100%)	Conformable Cable
			
Approximate Shielding Effectiveness Value			
-55dB	-90dB	-110dB	-150dB

Coaxial Cable Connector Types

Brass = ทองเหลือง

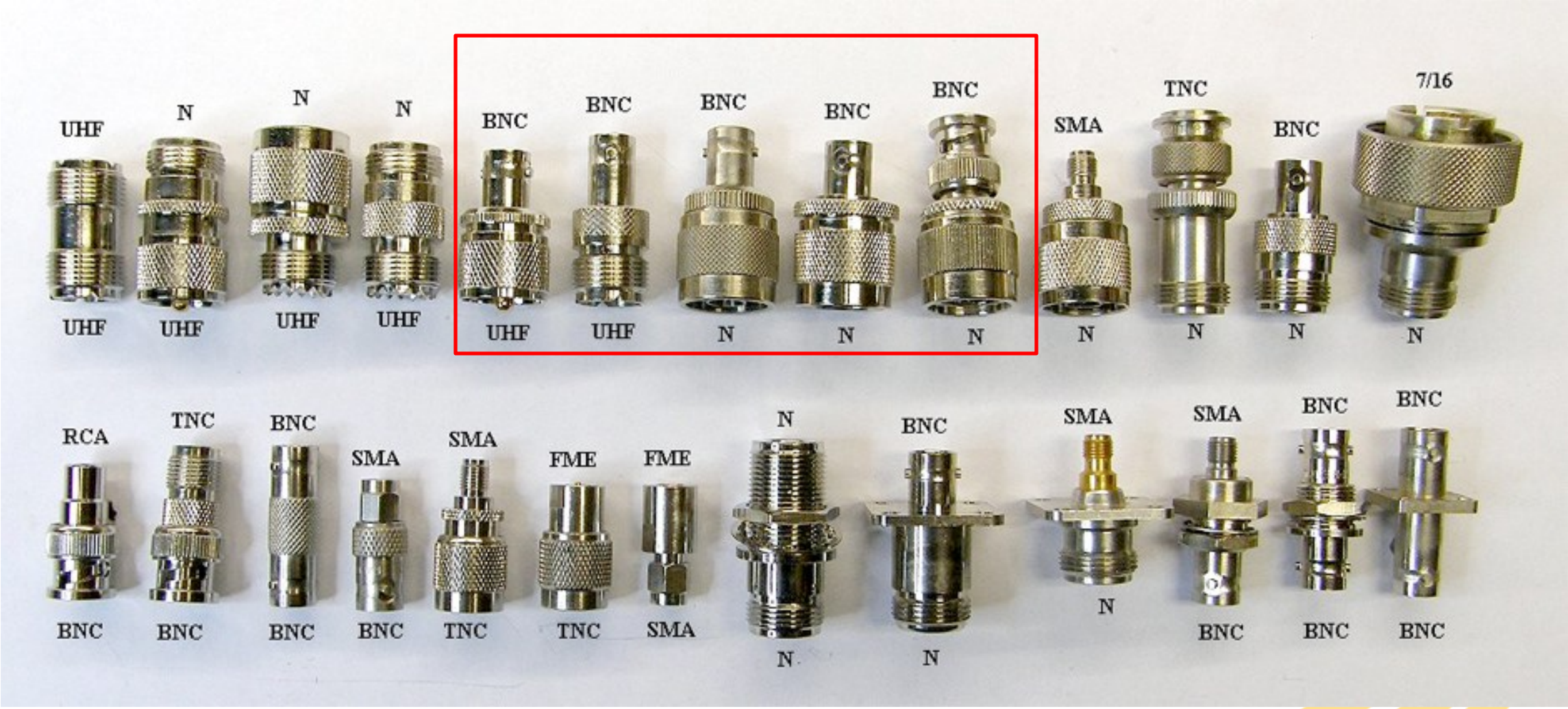
COAXIAL CONNECTORS AND THEIR FREQUENCY RANGES	
Connector type	Frequency range (GHz)
BNC	DC to 4
TNC	DC to 4
7-16 DIN	DC to 7.5
Type N	DC to 11
C	DC to 12
SMA	DC to 18
Precision SMA	DC to 26.5
3.5 mm	DC to 26.5
2.4 mm	DC to 50
2.92 mm/K	DC to 40
1.85 mm	DC to 65
1.00 mm	DC to 110



Different connector has different frequency range.

Coaxial Cable Connector Types

- Different types for different cables, applications and manufacturers



Coaxial Cable Connector Types

- Here are some popular ones.



Coaxial Cable Performance

nominal = minimal

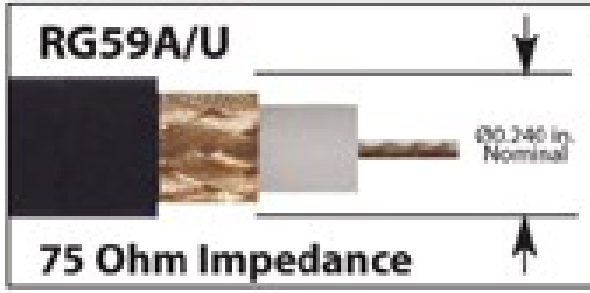
RG174/U



50 Ohm Impedance

NOMINAL ATTENUATION		
MHz	db/100 ft	db/100m
50	5.8	19.0
100	8.4	27.6
200	12.5	41.0
400	19.0	62.3
1000	34.0	111.5

RG59A/U



75 Ohm Impedance

NOMINAL ATTENUATION		
MHz	db/100 ft	db/100m
50	2.8	9.2
100	4.0	13.1
200	5.9	19.4
400	8.5	27.9
1000	13.8	45.3

RG316/U



50 Ohm Impedance

NOMINAL ATTENUATION		
MHz	db/100 ft	db/100m
50	5.6	18.4
100	8.3	27.2
200	12.0	39.4
400	17.5	57.4
1000	29.0	95.1

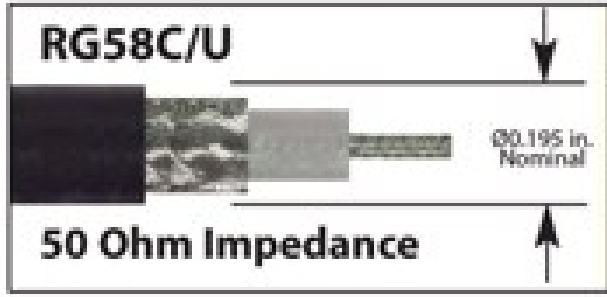
RG59B/U



75 Ohm Impedance

NOMINAL ATTENUATION		
MHz	db/100 ft	db/100m
50	2.4	7.9
100	3.4	11.1
200	4.9	16.1
400	7.0	23.0
1000	12.0	39.3

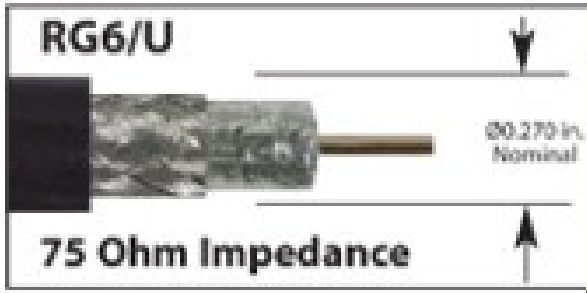
RG58C/U



50 Ohm Impedance

NOMINAL ATTENUATION		
MHz	db/100 ft	db/100m
50	3.3	10.8
100	4.9	16.1
200	7.3	23.9
400	11.0	36.1
1000	20.0	65.6

RG6/U



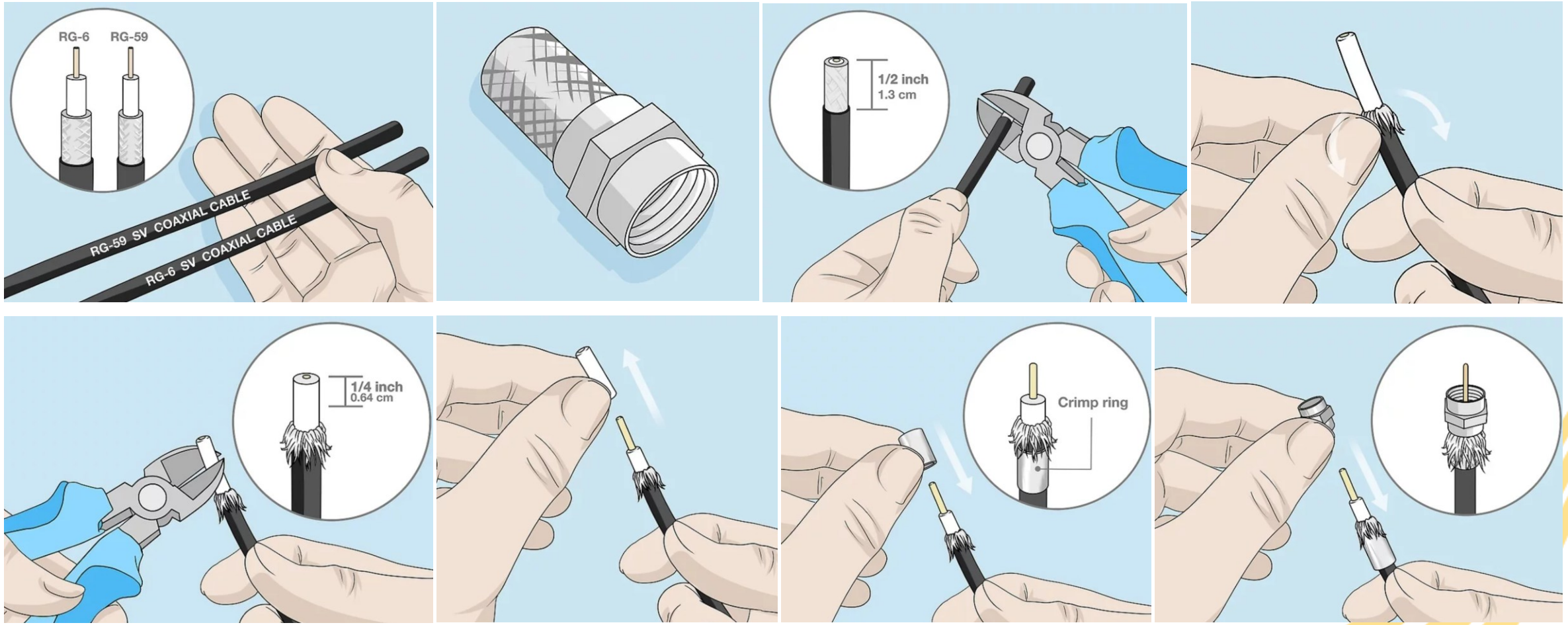
75 Ohm Impedance

NOMINAL ATTENUATION		
MHz	db/100 ft	db/100m
50	1.5	4.9
100	2.1	6.9
200	3.1	10.2
400	4.5	14.8
1000	7.3	23.9

Smaller attenuation, the better performance

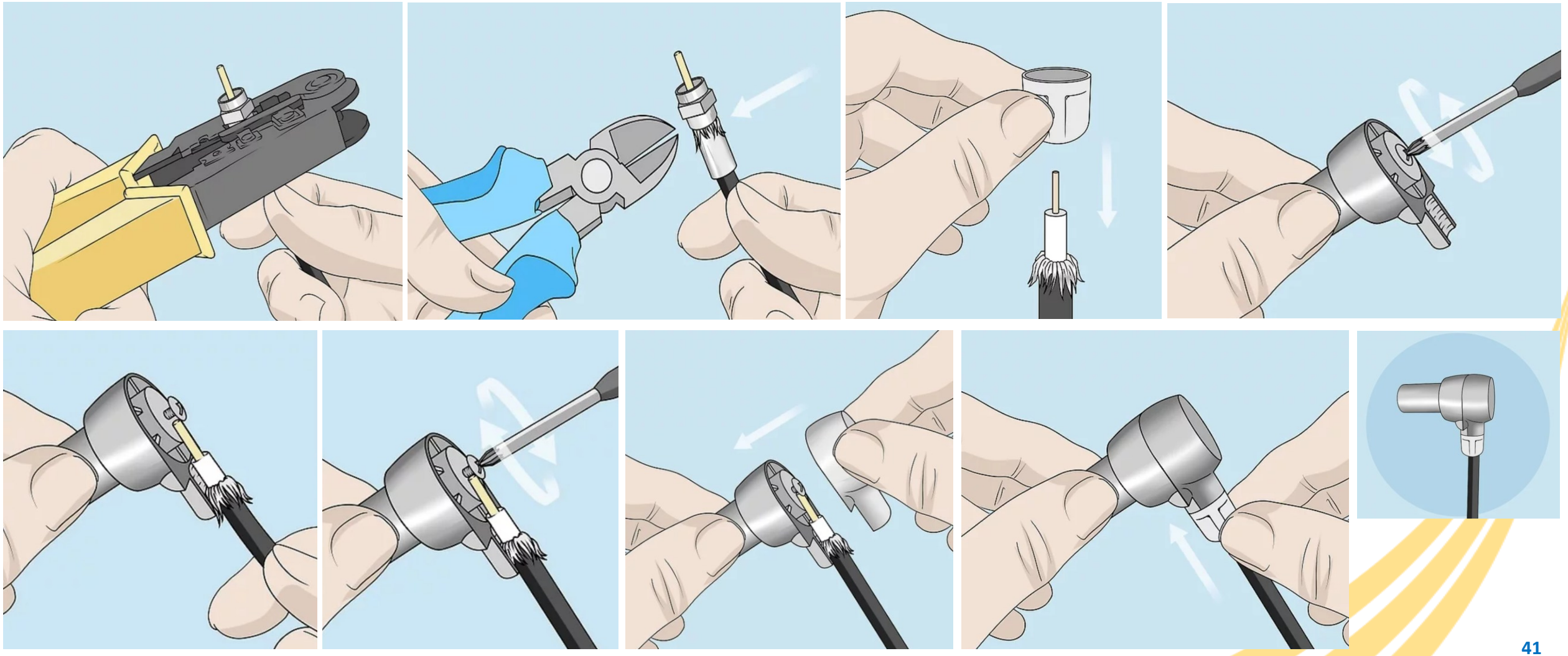
How to Attach Coaxial Cable Connectors

- Ref: https://www.wikihow.com/Connect-Coaxial-Cable-Connectors

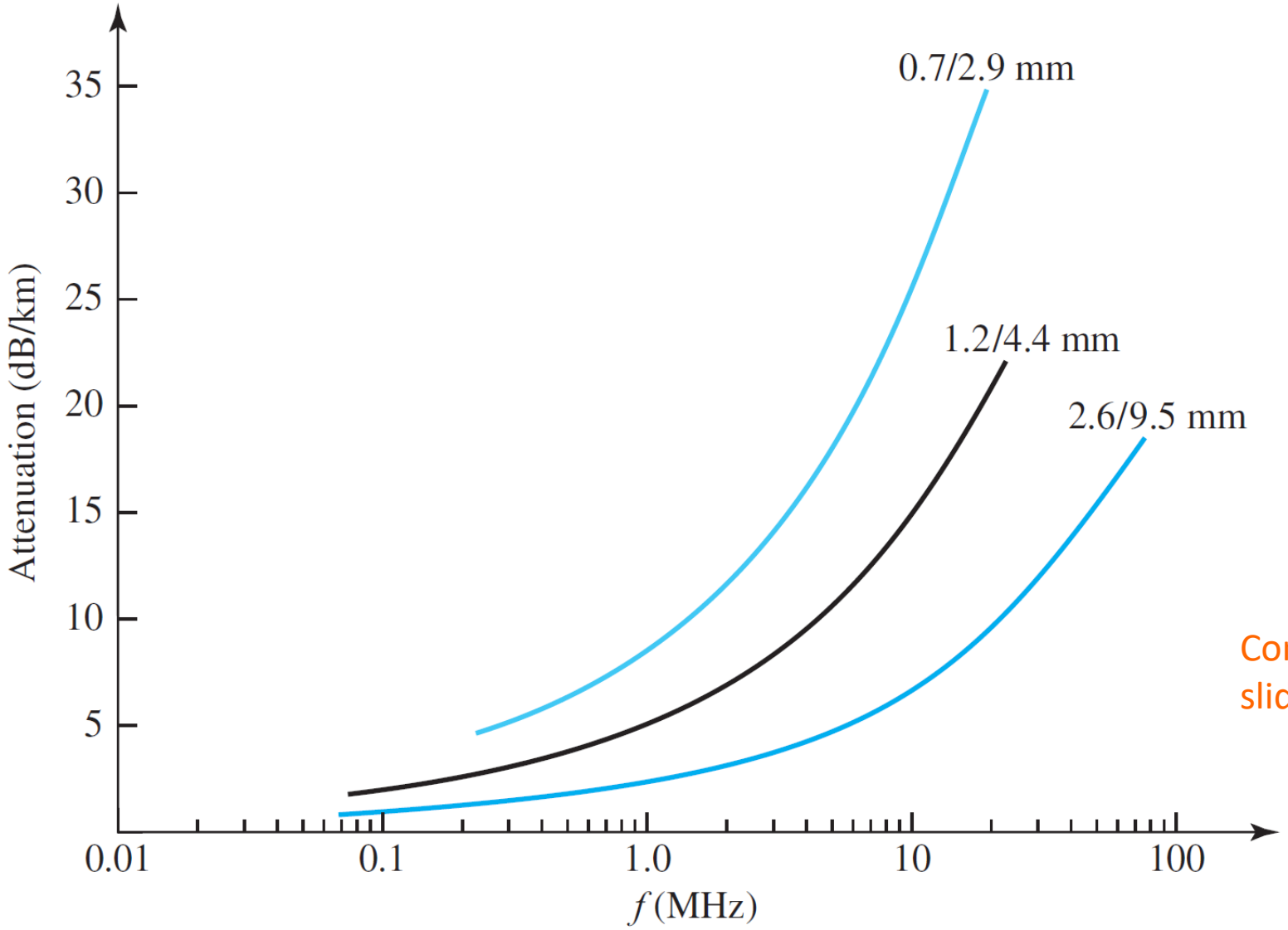


How to Attach Coaxial Cable Connectors

- Ref: https://www.wikihow.com/Connect-Coaxial-Cable-Connectors



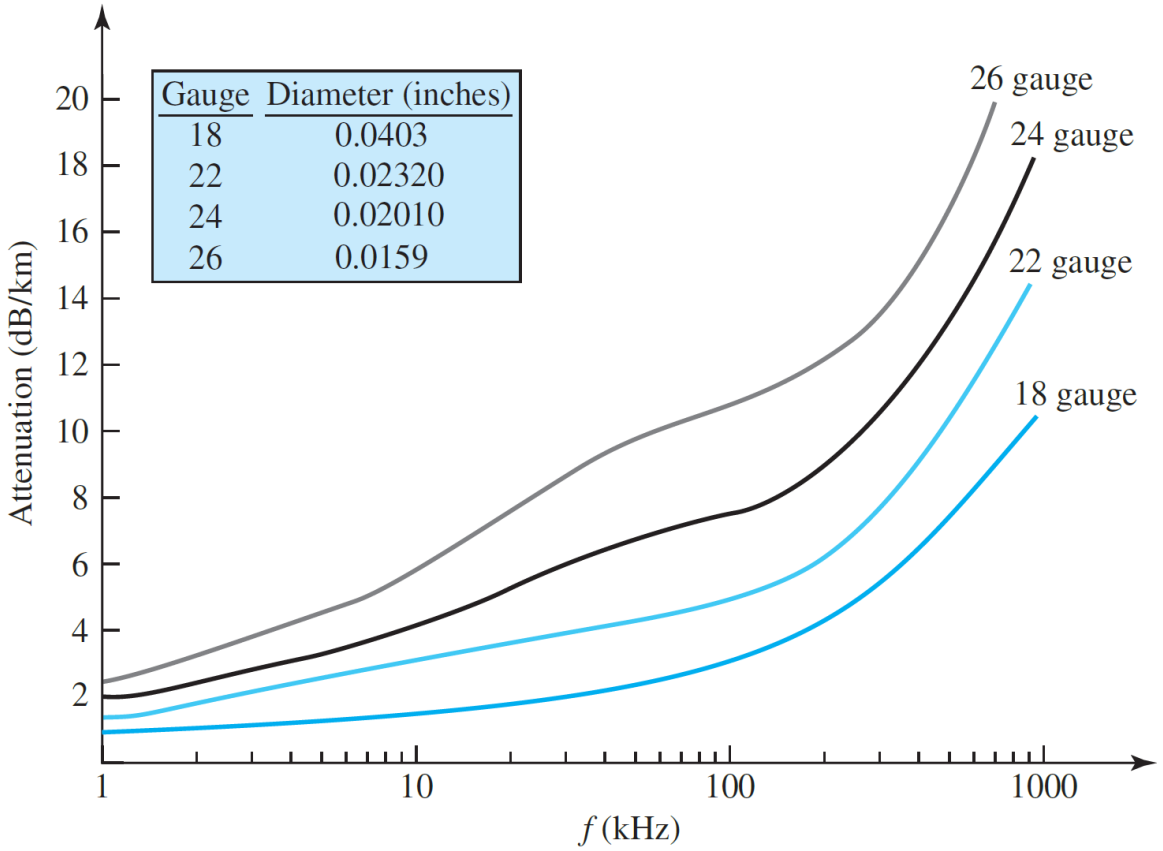
Coaxial Cable Performance



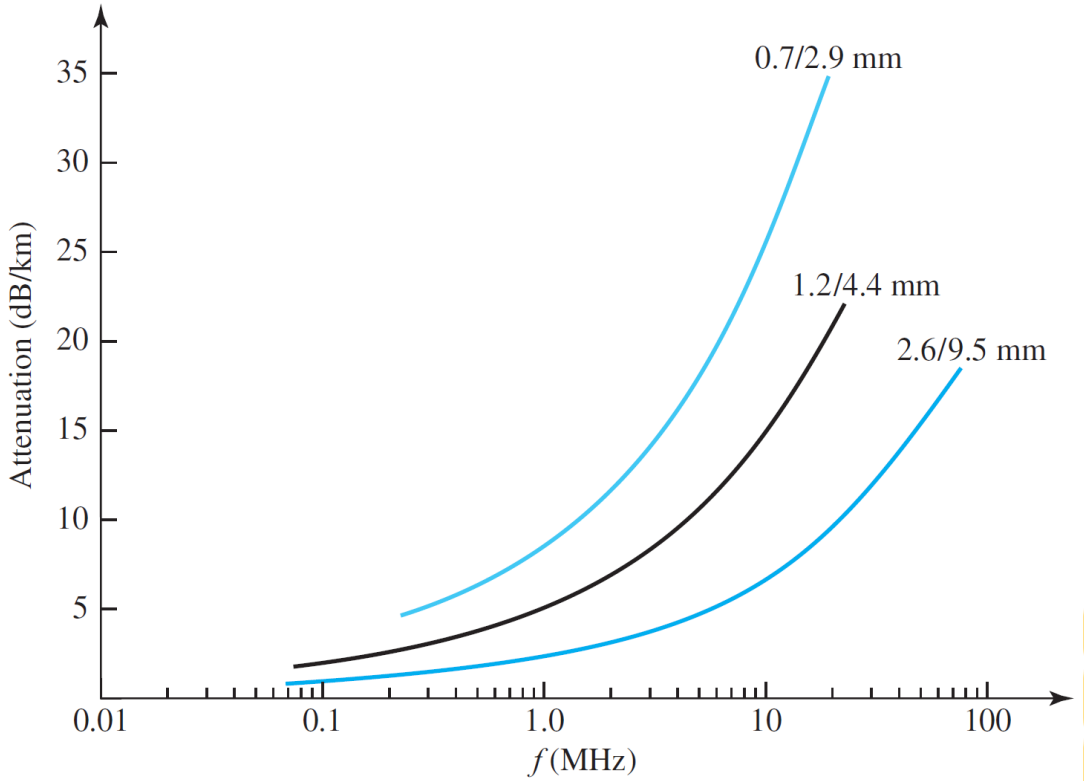
Compare this slide with slide 30

Coaxial Cable vs UTP Performance

UTP



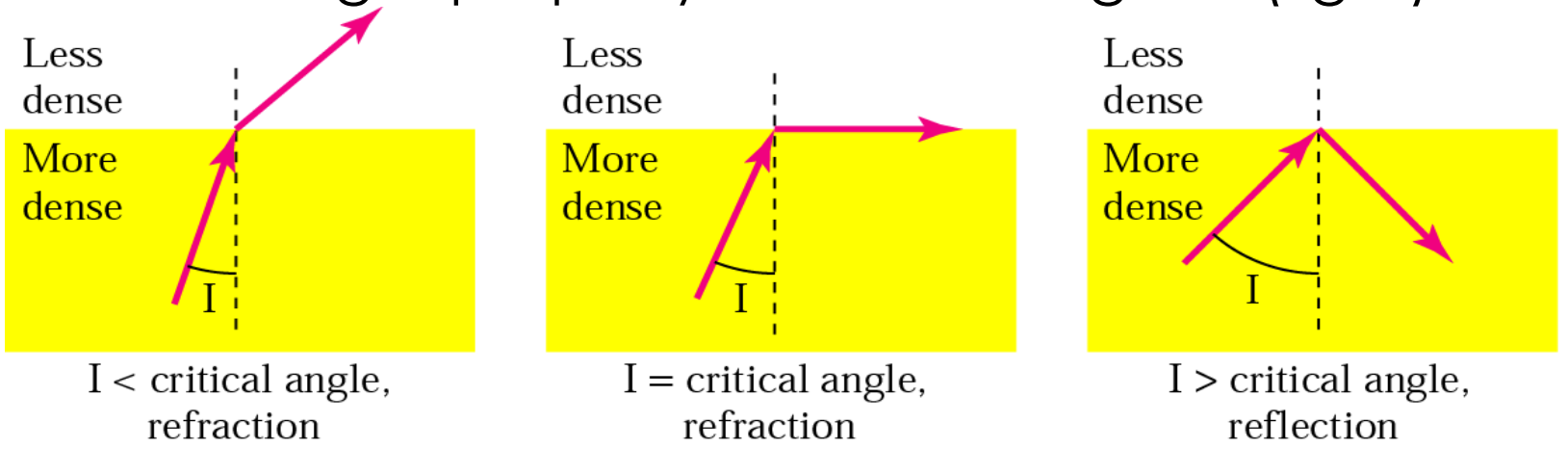
Coaxial

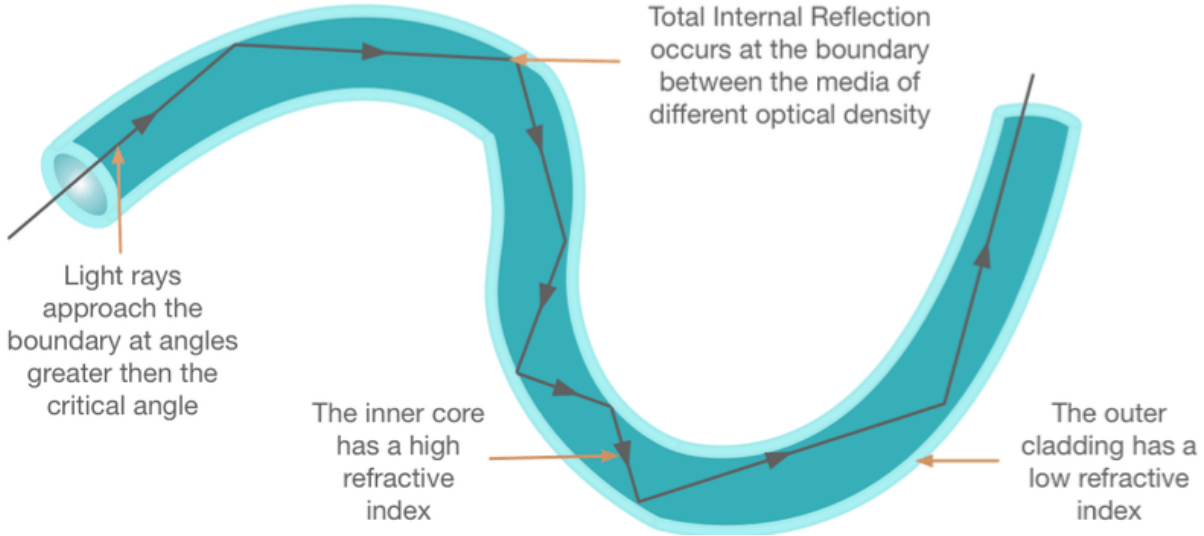


Coaxial cable outperforms UTP cable, especially in higher frequency.

Fiber-optic Cable

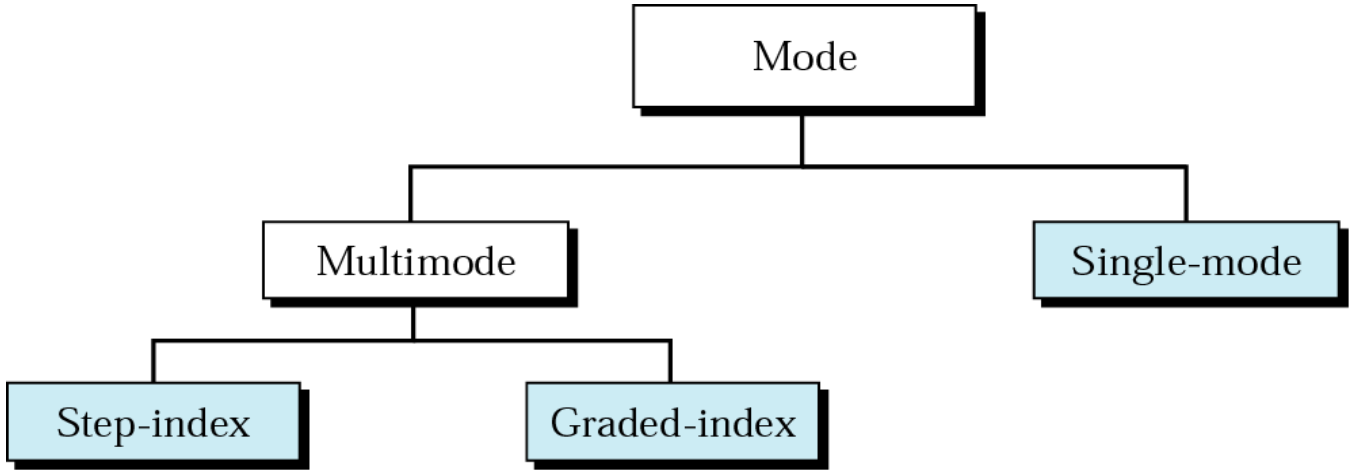
- Fiber-optic cable is made of glass or plastic which make use of the total internal reflection of light property to transmit signals (light).

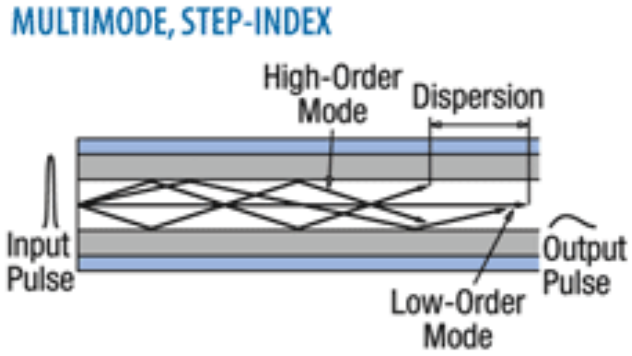


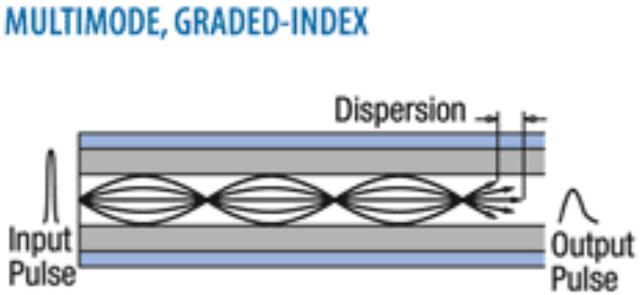


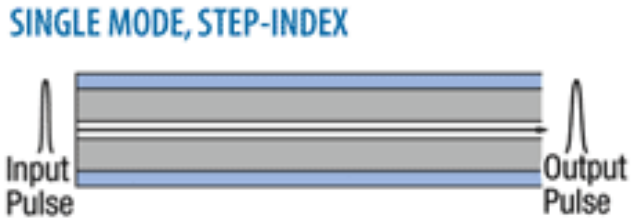
The core is surrounded by a cladding of less dense glass or plastic.

Fiber-optic Types





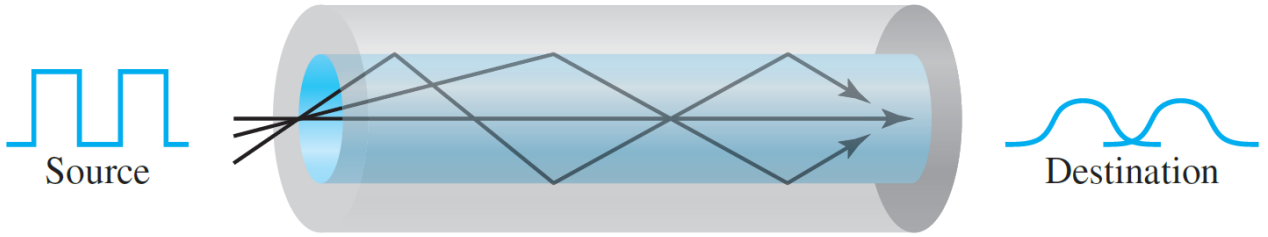




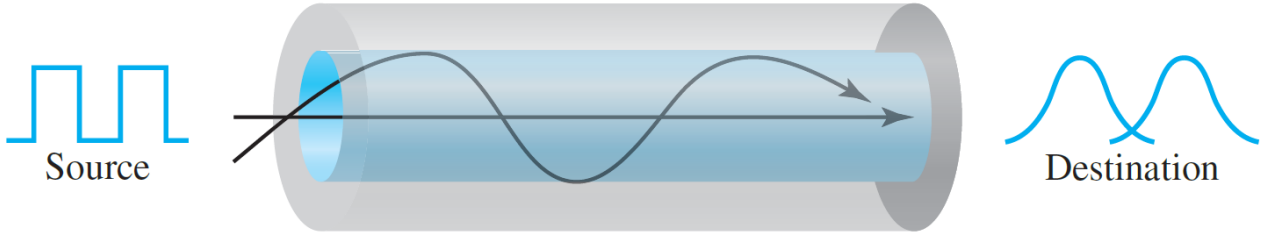
Fiber-optic Types

- There are two main types of fiber - single mode and multi-mode.
  - The **single-mode fiber**, which allows just one light source to pass along the cable, is typically used outside the datacenter for covering longer distances.
  - The **multi-mode version**, which allows multiple light sources down the cable for increased bandwidth, covers shorter distances and is ideally suited for the datacenter.
    - Fiber is graded using the Optical Multimode **(OM) standard**, and experts recommend the highest grade, **OM3**.

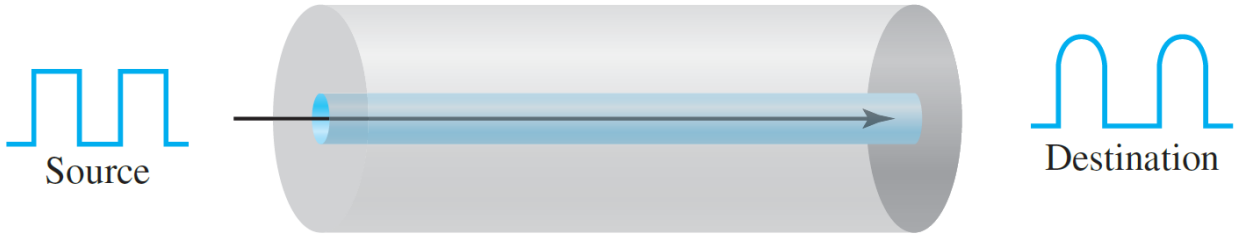
Fiber-optic Types



a. Multimode, step index



b. Multimode, graded index



c. Single mode

Fiber-optic Types

- Single-Mode
  - Uses step-index fiber and highly focused source of light that limits beams to a small range of angles, all close to the horizontal
  - The single-mode cable is manufactured with a much smaller diameter than the multimode fiber, with lower density.
  - Propagation of different beams is almost identical, and delays are negligible.
  - All the beams arrive at the destination "together" and can be recombined with little distortion to the signal

Fiber-optic Types

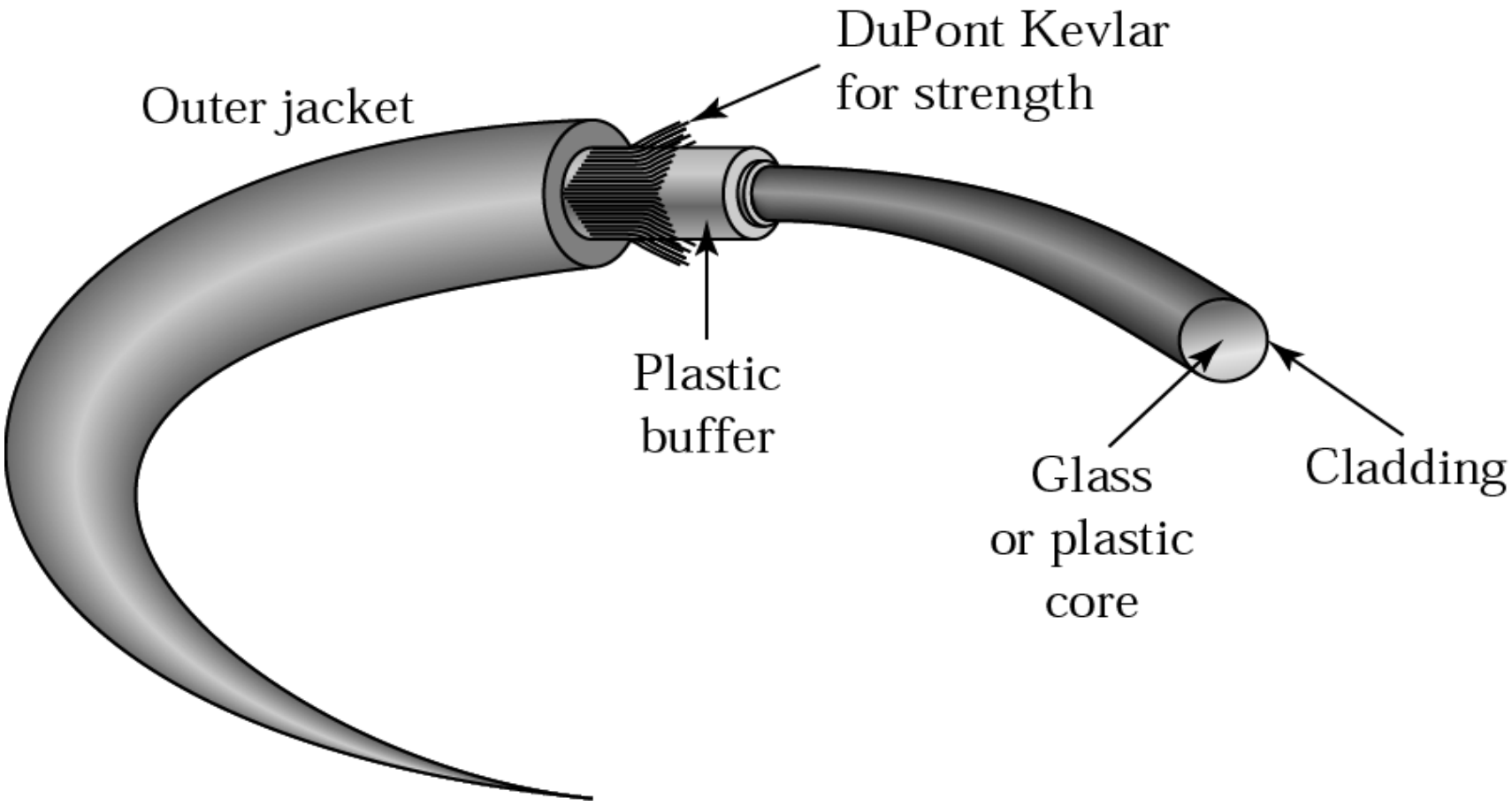
- Multi-mode step-index
  - The density of the core remains constant from the center to the edges.
  - A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding
  - At the interface, there is an abrupt change due to a lower density; this alters the angle of the beam's motion.
  - The term step-index refers to the suddenness of this change, which contributes to the distortion of the signal as it passes through the fiber.
- Multi-mode graded-index
  - decreases this distortion of the signal through the cable.
  - The word index here refers to the index of refraction. The index of refraction is related to density.
  - A graded-index fiber is one with varying densities. Density is highest at the center of the core and decreases gradually to its lowest at the edge.

Fiber-optic Types

- Plastic type
  - The core of the plastic-fiber consists of one or more acrylic-resin fibers 0.25 to 1 mm 0.01" to 0.04" in diameter, encased in a polyethylene sheath.
  - Plastic fibers are light, cost-effective, and flexible which is why they are the most common type of fiber sensor.

- Glass type
  - The glass-fiber consists of 10 to 100 µm 0.39 to 3.94 Mil diameter glass fibers encased in stainless steel tubing.
  - This allows it to be used at high operating temperatures (350°C or 662°F max.).

Fiber-optic Protection

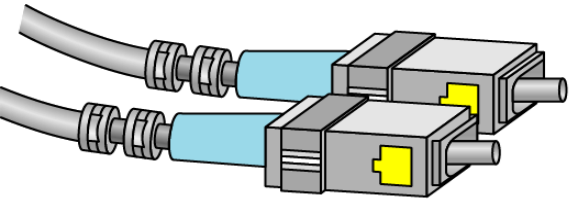


Fiber-optic Types

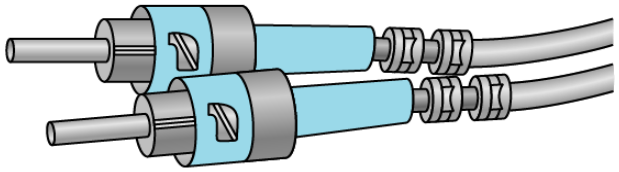
- Optical fibers are defined by the ratio of the diameter of their core to the diameter of their cladding (in micrometers)

Type	Core	Cladding	Mode
50/125	50	125	Multimode, graded-index
62.5/125	62.5	125	Multimode, graded-index
100/125	100	125	Multimode, graded-index
7/125	7	125	Single-mode

Fiber-optic Connector Types



SC connector



ST connector



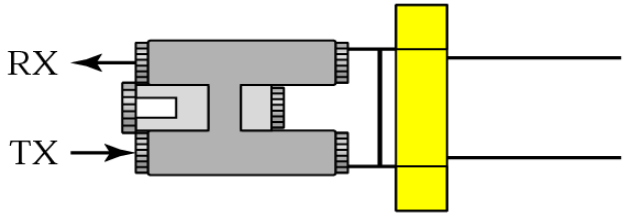
LC Connector



SC Connector



ST Connector



MT-RJ connector



MTRJ



MU



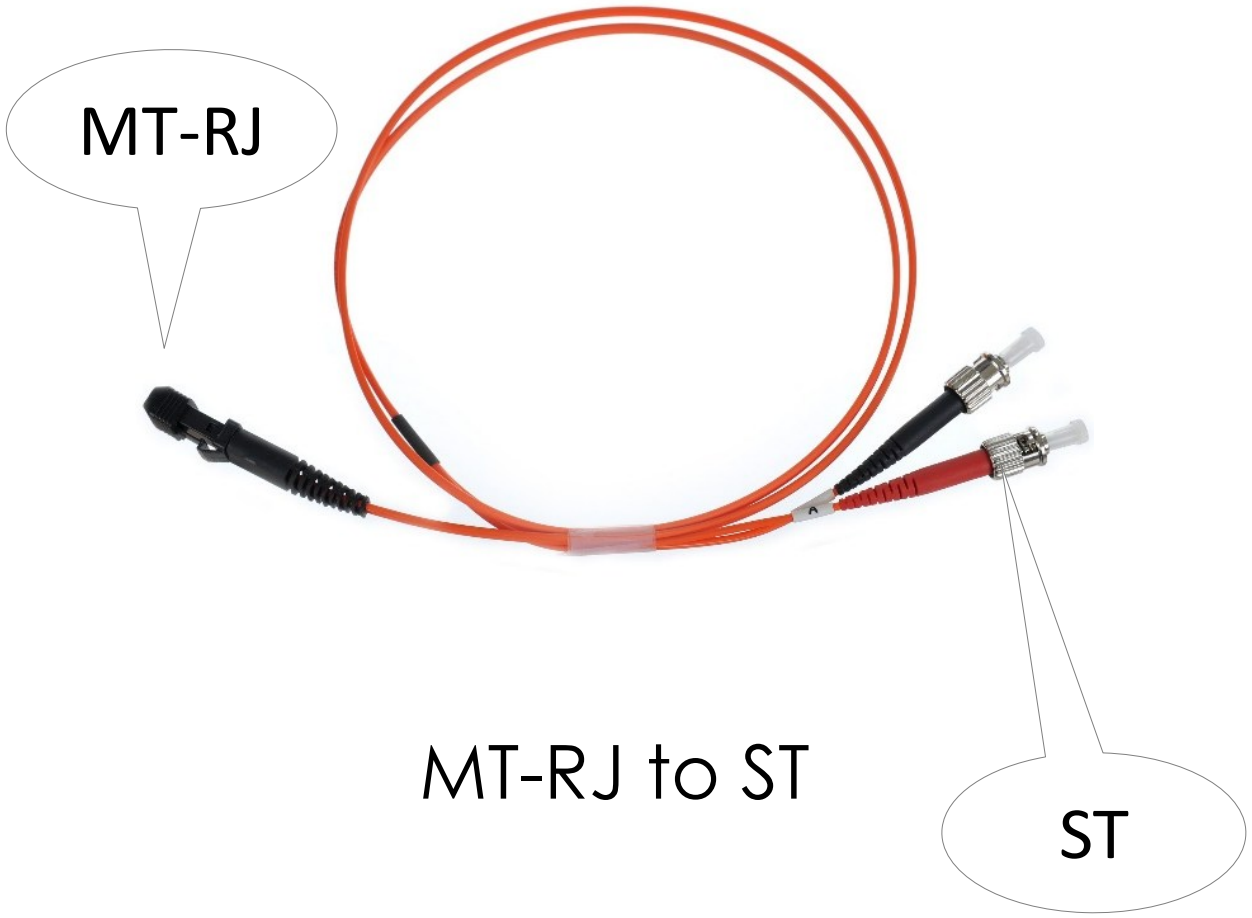
E 2000 Connector

Fiber-optic Connector Types

	Connector Type	Coupling Type	Fiber Type	Polish	No. of Fibers	Typical Applications	Comment
	ST	Twist on	Single mode /Multimode	PC, UPC	1	LANs	Keyed
	FC	Screw on	Single mode /Multimode	PC, UPC, APC	1	Datacom, Telecommuni-cations	Keyed
	SC	Snap on	Single mode /Multimode	PC, UPC, APC	1	CATV, Test Equipment	Keyed
	LC	Snap on RJ45 style	Single mode /Multimode	PC, UPC, APC	1	Gigabit Ethernet, Video Multimedia	Small Form Factor (SFF)
	MU	Push/Pull	Single mode /Multimode	PC, UPC, APC	1	Medical, Military	Small Form Factor (SFF)
	MT-RJ	Snap on RJ45 style	Single mode /Multimode	N/A	2	Gigabit Ethernet, Asynchronous Transmission Mode (ATM)	One of Mating Connectors must have Alignment Pins
	MPO (MTP)	Push/Pull	Single mode /Multimode	N/A	4, 8, 12, 16, 24	Active Device Transceiver, Interconnec-tions for O/E Modules	One of Mating Connectors must have Alignment Pins

Fiber-optic Converter Cables

- There are cables and connectors that can change one connector type to another



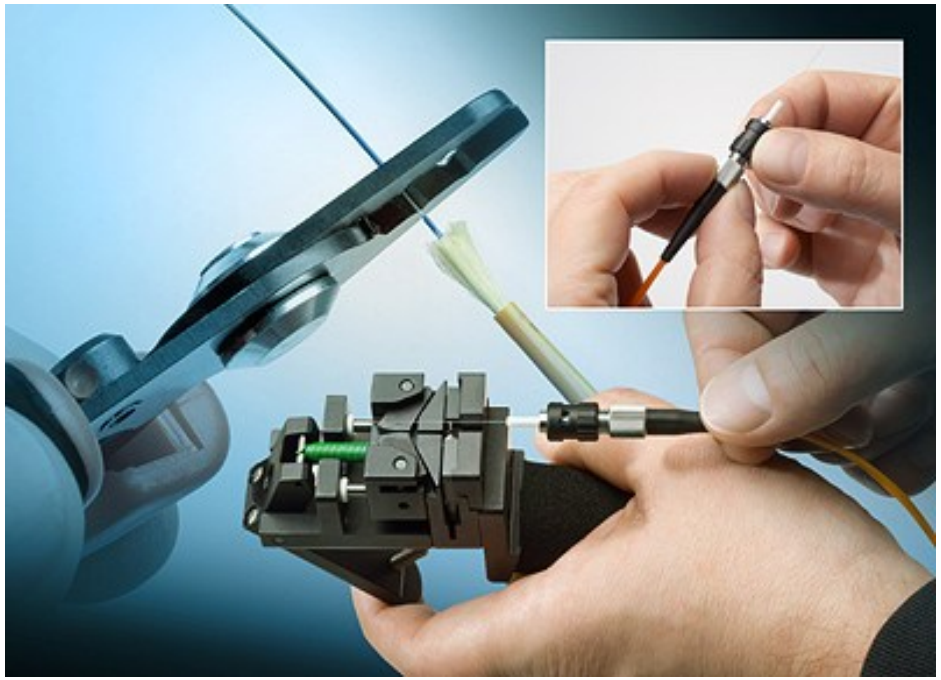
MT-RJ to ST

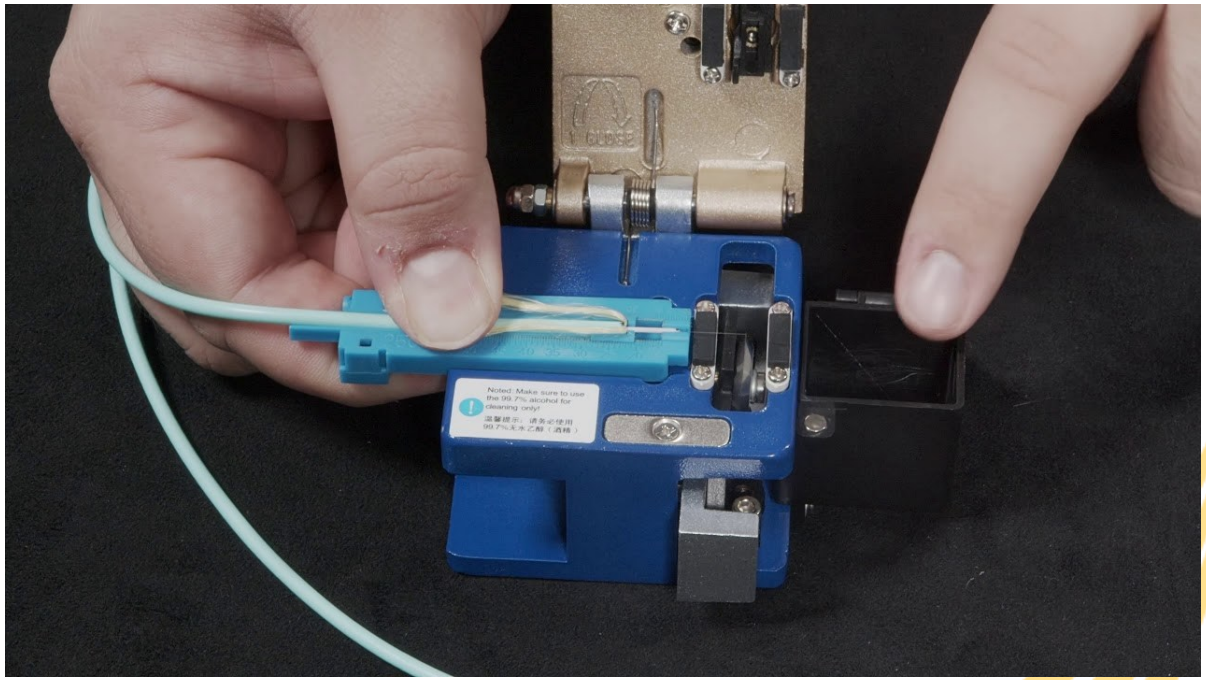


MT-RJ to SC

Terminating Fiber Optic Cable

- Terminating fiber optic cable requires a special tool





Fiber-optic Tester



Measure signal quality



Find broken point

Fiber-optic Cable Maintenance

- Cleaning the head







Fiber Optic Maintenance

- Attaching the jacks, cutting and fusing the cable
- https://www.youtube.com/watch?v=TPVvb6cmjds
- https://www.youtube.com/watch?v=ekzlonBS7d8

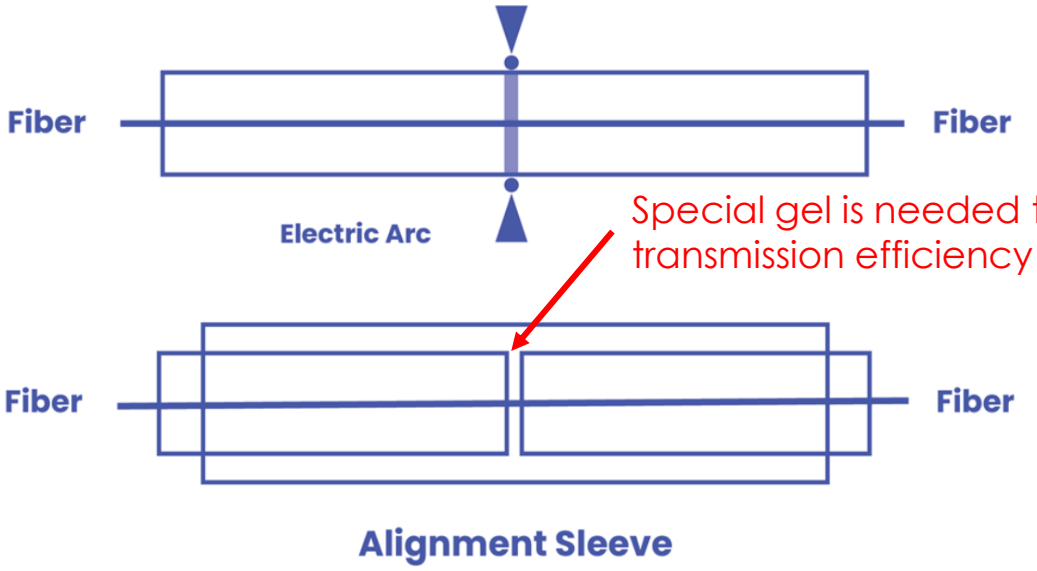


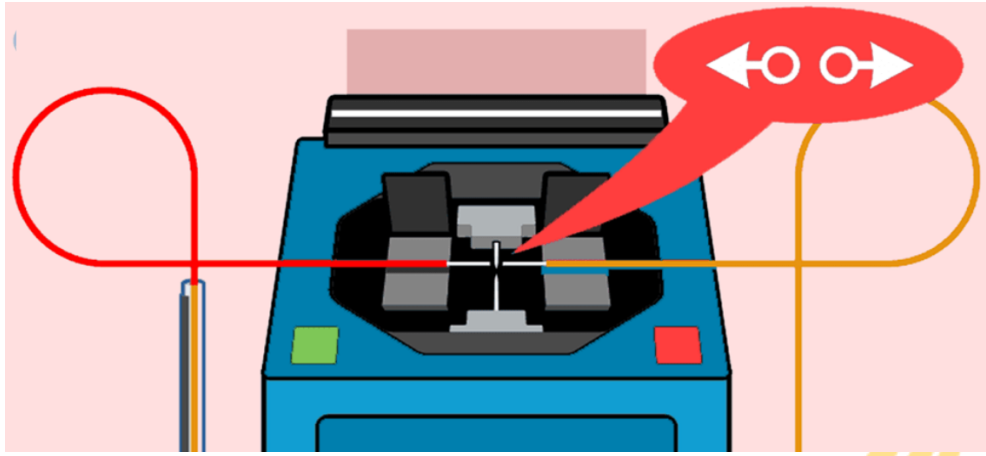




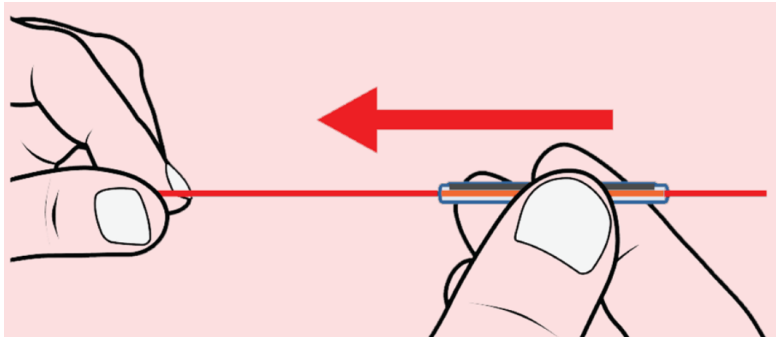
Fiber Optic Maintenance

- Reference:
- https://www.cepro.com/audio-video/cabling/step-by-step-guide-to-fiber-optic-cable-splicing-for-integrators/





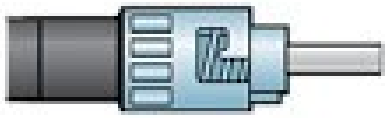
Fusing cables



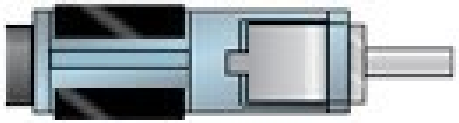
fusion splicing

Summary of Fiber Connectors

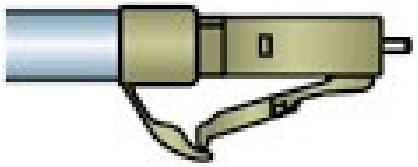
- Summary of connectors used with network equipment



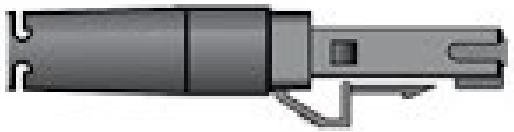
ST Connector



SC Connector



LC Connector



MT-RJ Connector



RJ45 Connector

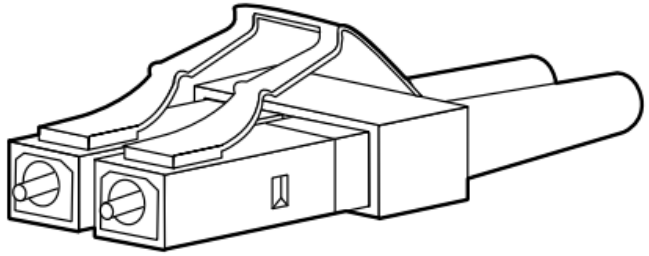


SFP,SFP+,XFP transceivers

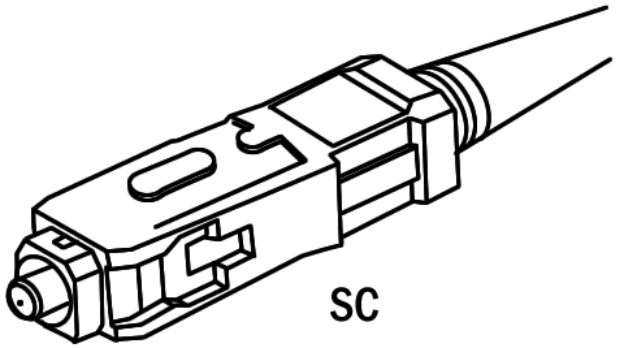
Cables for Cisco

- **Copper cable**
  - RJ-45 with CAT 5e or CAT 6 are commonly used for copper-based cables supporting up to 1 Gbps

- **Fiber-optic cable**
  - Cisco switches and routers support many types of connectors (jacks)
    - 100BASE-FX ports use MT-RJ connectors with 50/125 or 62.5/125 micron multimode fiber-optic cabling.
    - 1/10 Gbps ports usually use SFP modules with LC connectors (secure lock) or SC connectors



LC



SC

Fiber-optic Transceiver

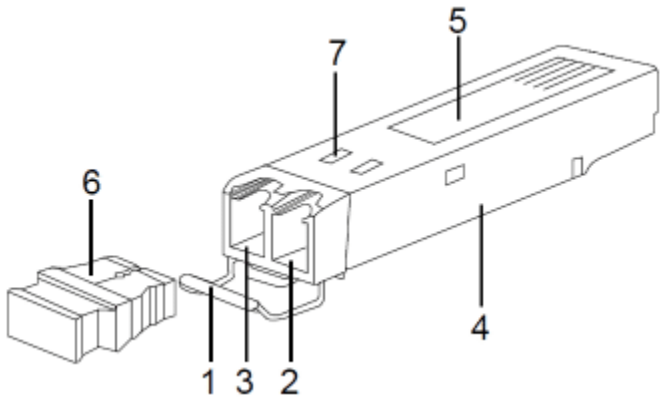
- Fiber optic transceiver is a device that uses fiber optical technology to send and receive data.
- **The transceiver encodes/decodes/converts electrical signals into light pulses** and then sends them to the other end as electrical signals.

- Optical fibers are classified into the single mode fiber SMF and Multi-mode fiber MMF. Therefore, fiber optic transceiver is also classified as **single-mode and multi-mode transceivers** to support different optical fibers. The single-mode transceivers usually have a transmission distance of 10km to 160 km, while the multi-mode one's used for short transmission distance such as 0.5km to 2km.

- Reference: https://www.optcore.net/what-is-fiber-optic-transceiver/

Fiber-optic Transceiver

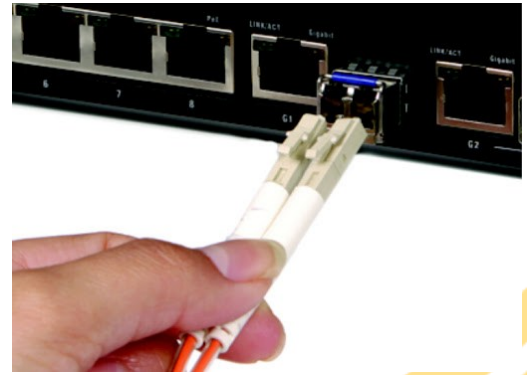
- Most fiber-optic transceivers look like this



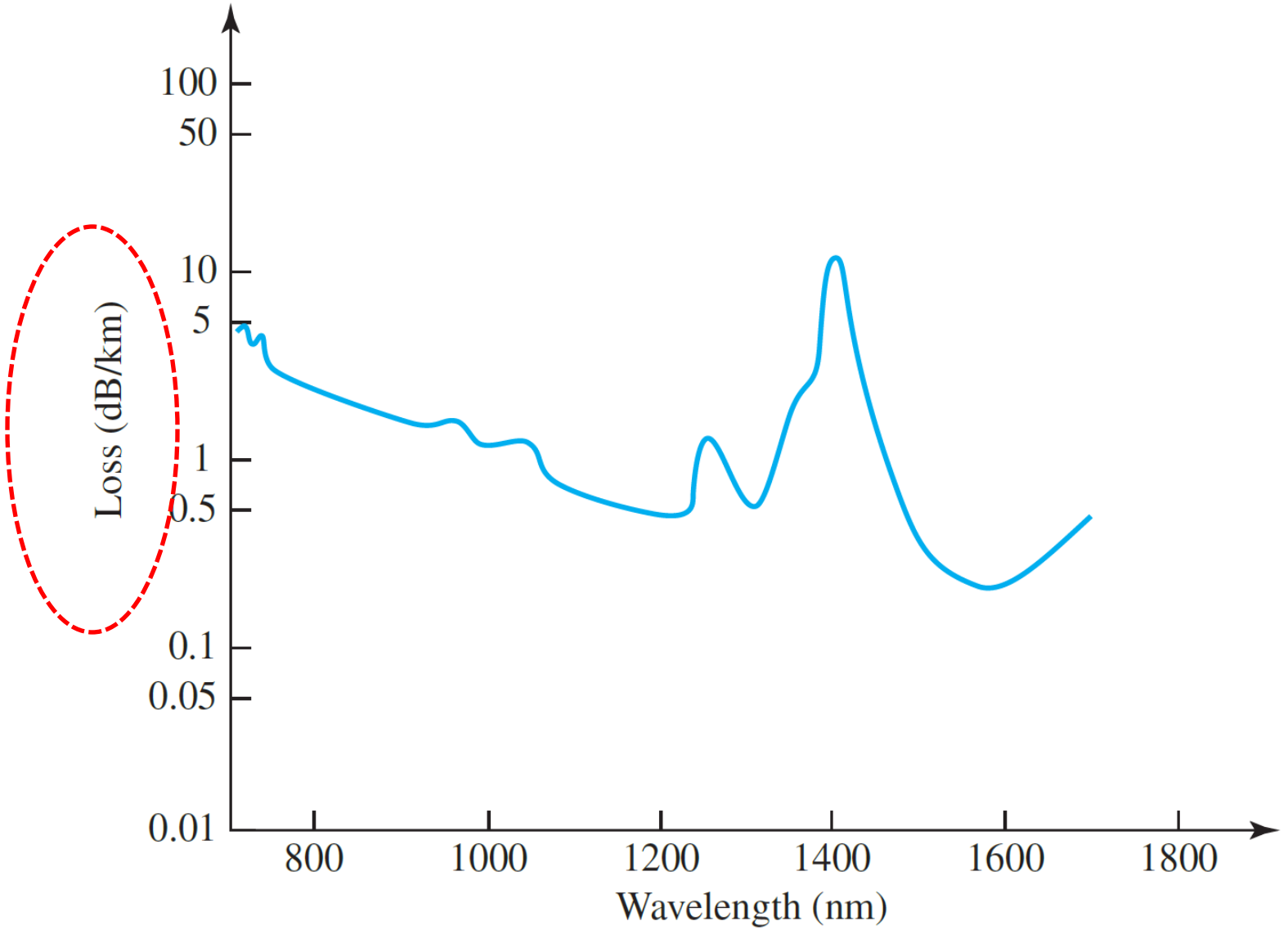




1	Latch
2	Optical Receiver
3	Optical Transmitter
4	Transceiver Shell
5	Label
6	Dust Plug
7	Spring

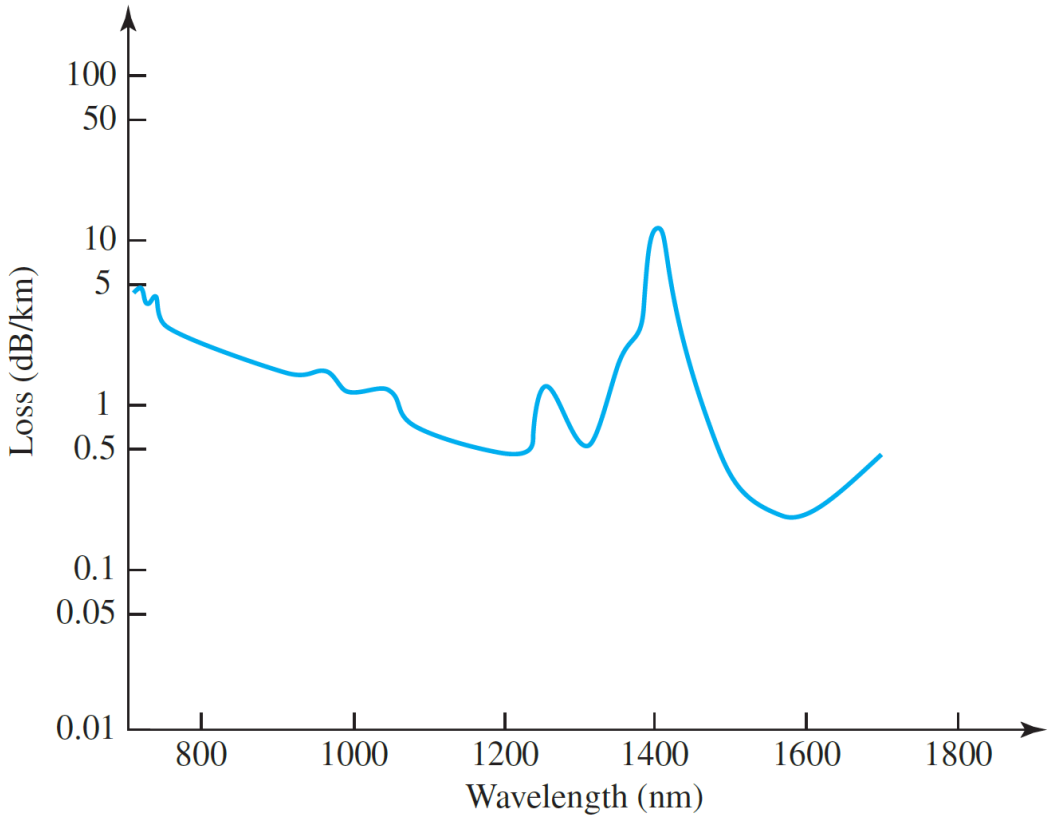


Fiber-optic Performance

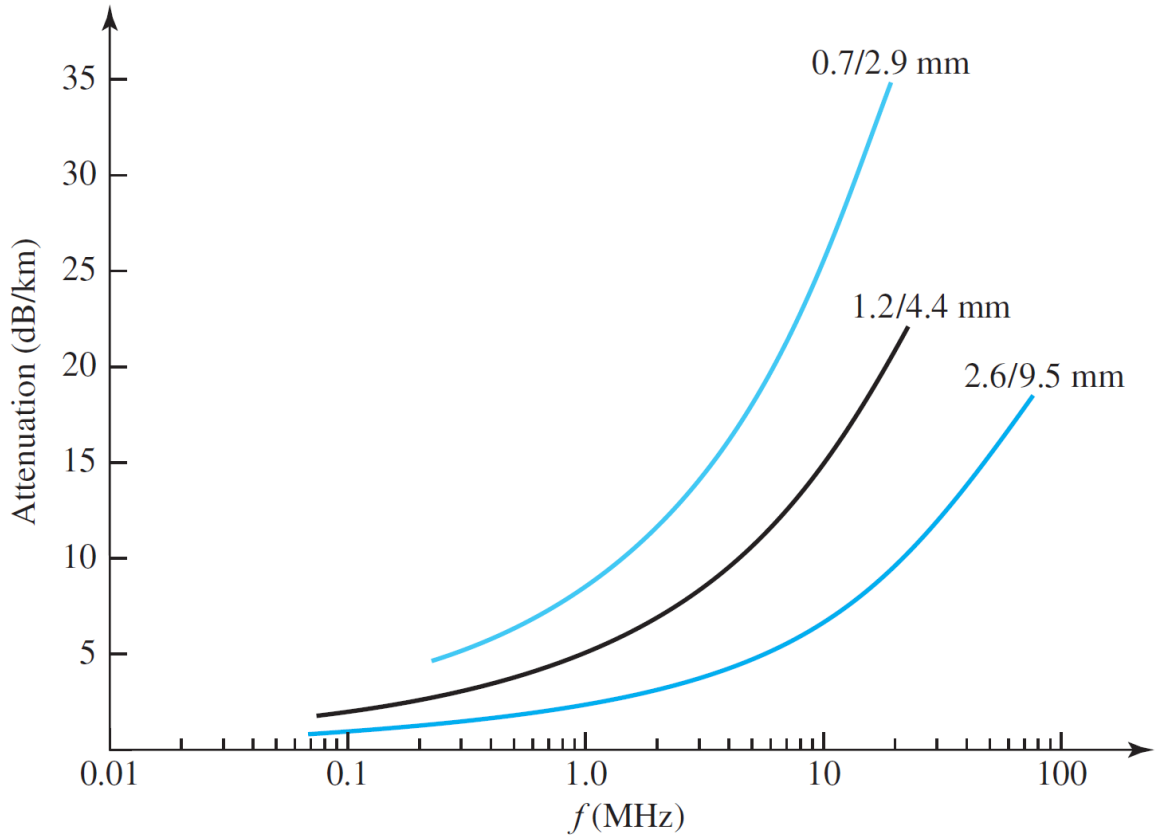


Compare this slide with slide 42

Fiber-optic Performance



Fiber optic



Coaxial

Advantages and Disadvantages of Fiber cables

- **Advantages**
  - Higher bandwidth
  - Less signal attenuation
  - Immunity to electromagnetic interference (EMI)
  - Resistance to corrosive materials
  - Light weight
  - Greater immunity to tapping
- **Disadvantages**
  - Installation and maintenance
  - Unidirectional light propagation
  - Cost

Commonly Used Fiber Optic Cables

- Fiber optic cables support much of the world's Internet and television connectivity.

- Single-mode fiber (SMF) carries only a single beam of light. This is more reliable and supports much higher bandwidth and longer distances than the MMF cable. This cable uses a laser as the light source and transmits 1300 or 1550 nanometer wavelengths of light.

- Multi-mode fiber (MMF) carries multiple beams of light. Because of multiple beams, this cable carries much more data than the SMF cable. This cable is used in shorter distances. This cable uses an LED as the light source and transmits 850 or 1300 nano-meter wavelengths of light.

Fiber optics under deep sea

- The fight over the internet, under the sea | CNBC Explains (with shark) **
- https://youtu.be/8wa8q8wRZyU



- Meet Curie, Google's international fiber optic subsea cable
- https://youtu.be/ltE37wwbU9g

- Submarine Cable installed & Maintenance Archives
- https://youtu.be/s5SZDfRSwBo

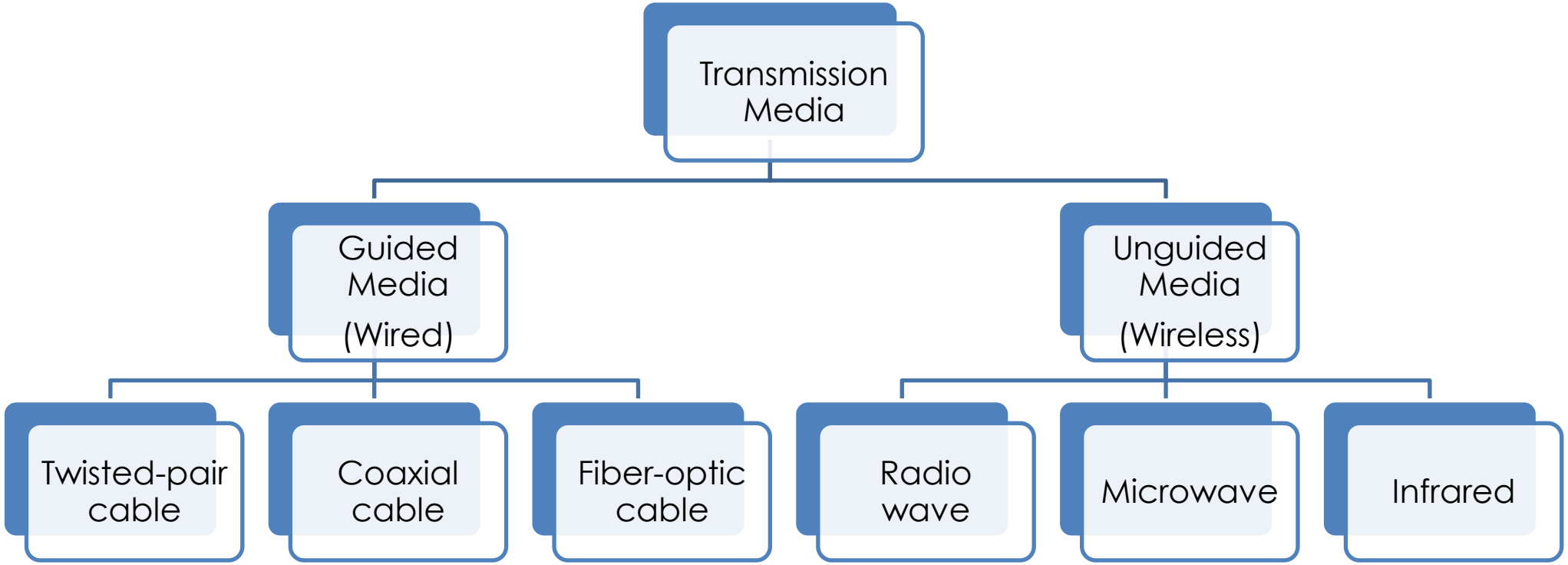
- How a Few Undersea Cables Connect the Entire Internet
- https://youtu.be/eTBLIYJSzdc



- How The Internet Travels Across Oceans ***
- https://youtu.be/yd1JhZzoS6A

- Shark attacks on submarine cable
- https://youtu.be/1ex7uTQf4bQ

Transmission Media



Cable standard	CAT	RG	SMF, MMF diameter of core/ diameter of cladding
Connector standard	RJ	BNC, N, UHF, etc.	Connector: MT-RJ, SC, ST, etc.